FALCON
AUTOTECH

FALCON’s Response to Prime IQ for Supply of Loop CBS 24K PPH

October 30, 2023

www.falconautotech.com

Kind Attention –

Mr. Lateef

M/s Prime IQ

Offer Ref: F23-00239; Date: 30-10-2023

Subject – Techno -Commercial Offer of Loop Cross Belt Sorter 24K PPH

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have studied your requirements in great depth and, along with the information collected during the meetings and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references.

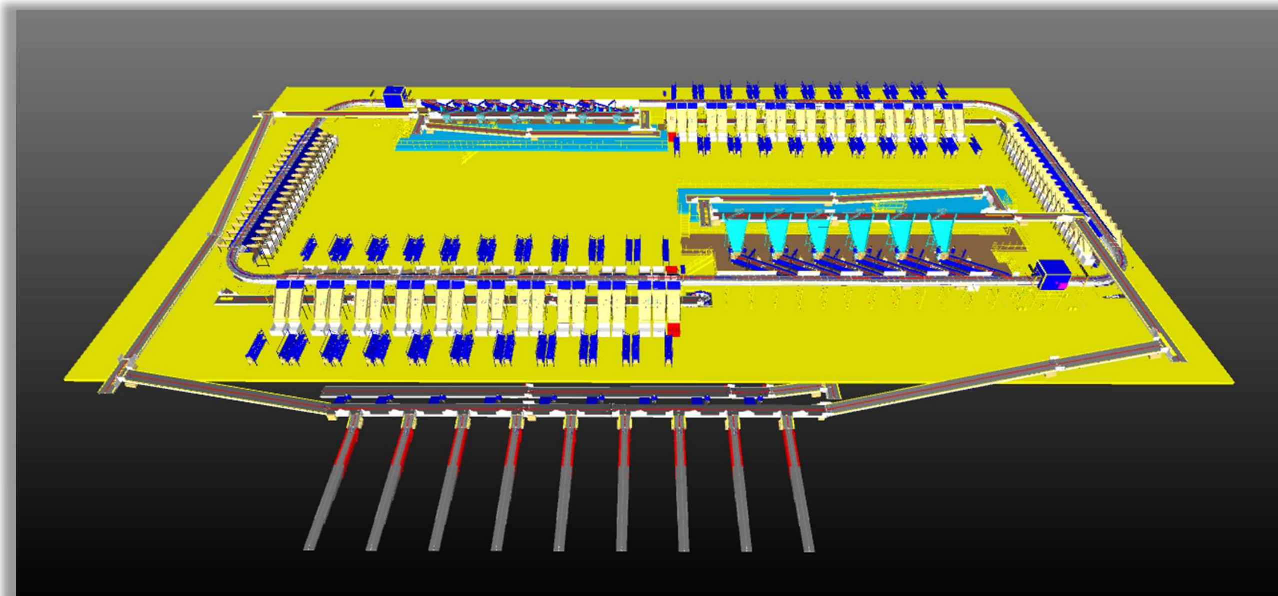
To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

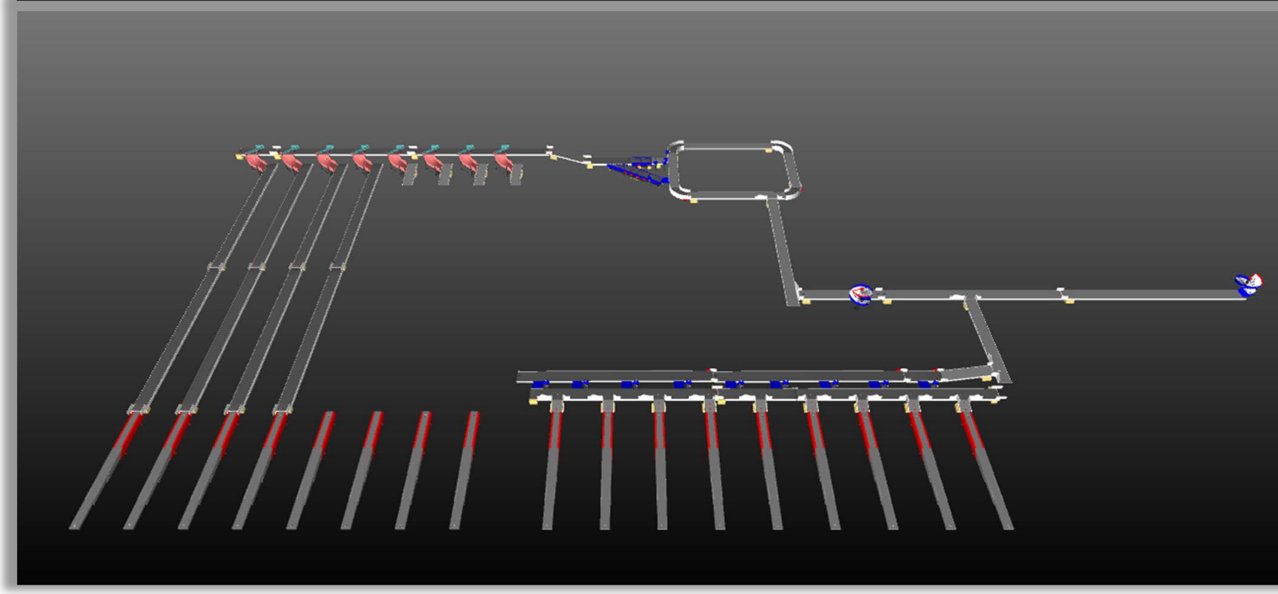
Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFP for Loop Cross Belt Sorter 24K PPH Throughput





Proposal Reference – F23-00239

Contents

1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Shipments Per Hour
4	ICR	Intelligent Character Recognition
5	MEZZ	Mezzanine
6	LIM	Linear Induction Motors
7	ECDS	Empty Carrier Detection System
8	AC	Alternating Current
9	DC	Direct Current
10	PLC	Programmable Logic Controller
11	IT	Information Technology
12	BOQ	Bill Of Quantity
13	I/O	Input/ Output
14	PDP	Power Distribution Panel
15	PC	Personal Computer
16	UPS	Uninterrupted Power Supply
17	CBS	Cross Belt Sorter
19	MDR	Motor Driven Roller
20	DAP	Design Approval Phase

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this Prime IQ RFQ of Loop CBS for Baghdad. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success. Following the same objective for the Baghdad CBS system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **One loops of shipment sorting, employing our cross-belt technology, have been designed to handle a throughput of 20,000 packages per hour.**
- **The loop is equipped with its own infeed conveyor system, comprising two distinct infeed lines.**
- **Within the loop there are two induction zones featuring six induct lines.**
- **In each zone, there are 40 Direct bagging chutes, 40 secondary chutes (Collection Type) and 2 rejection chutes.**
- **The system layout has been meticulously planned to facilitate smooth operational flow, ensuring efficient movement of personnel.**
- **Furthermore, additional conveyor automation is implemented to take out the bags (containing sorted parcel) & connecting it to Outbound sorter's infeed present at lower level with the help of Spiral chutes.**
- **Lastly, Outbound bag sorter is present at ground level followed by conveyor automation.**

A tailor-made and simple layout designed explicitly for Prime IQ. The proposed layout is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our meeting.

1. Falcon's reliable Shipment sortation systems

These systems are globally being used by most innovative brands such as Amazon, Delhivery, Asendia, Fastway, Aramex and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

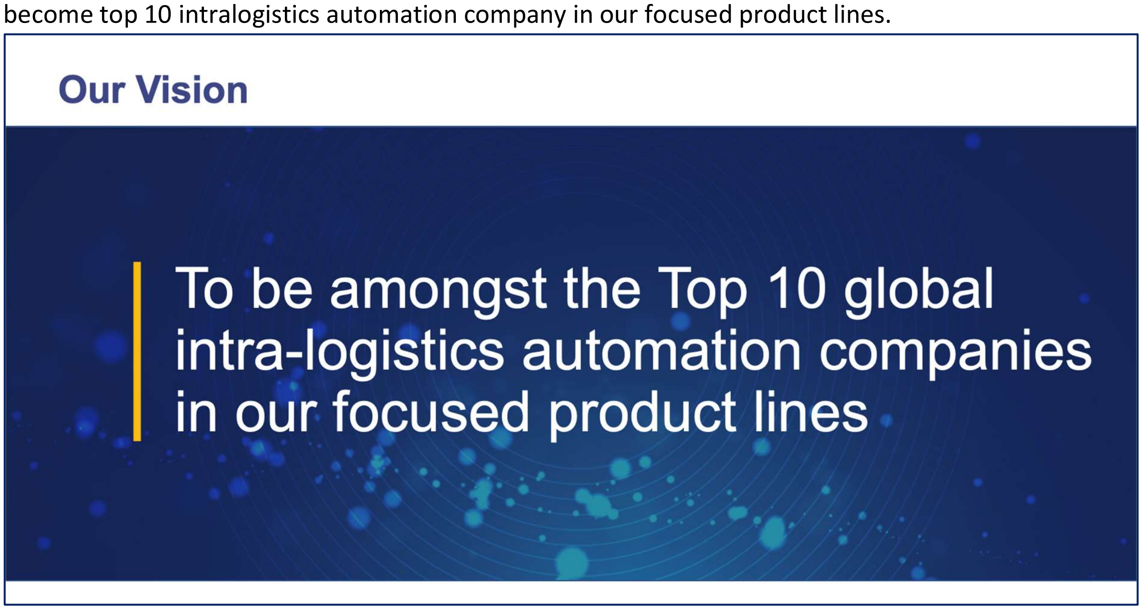
2. Commitment to quality systems

Demonstrating Falcon's clear commitment to the Prime IQ satisfaction, the shipment sortation system, parts and services will be under warranty for 12 months from installation go-live.

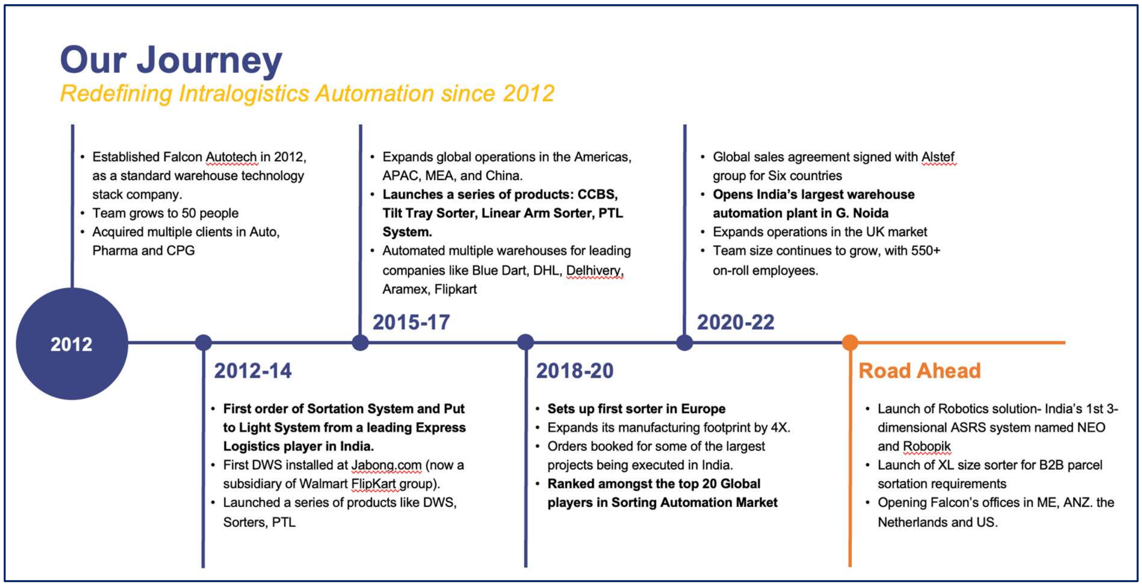
3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

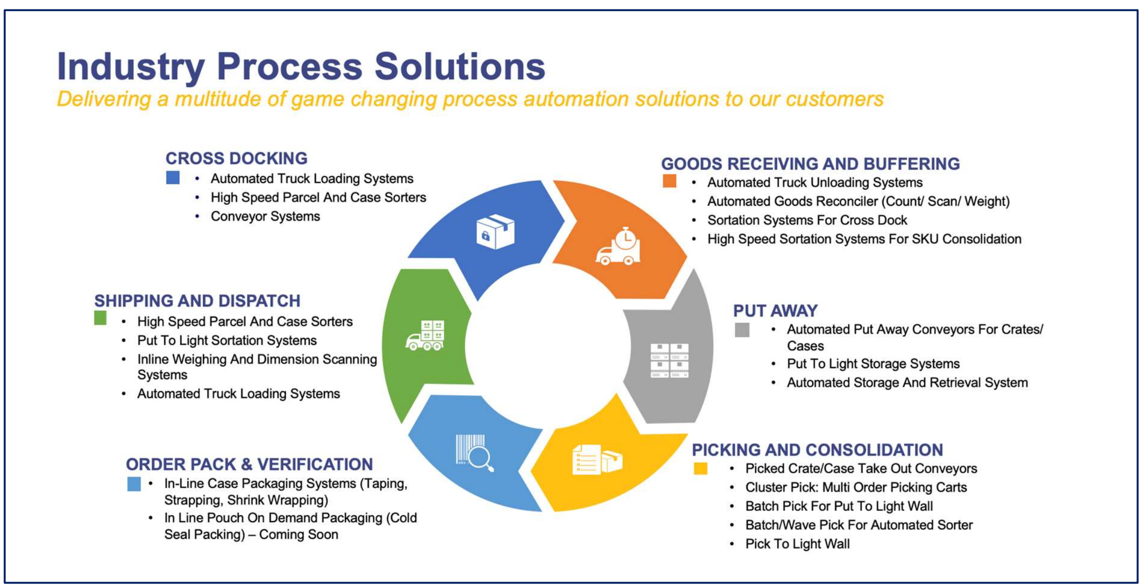
Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chai

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –

5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of shipments arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Shipment handling.
- FOCR Engine.
- WCS Software System.

Site Picture –

5.3 Project 3- (Client – Courier Express Shipment, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 shipments per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –

5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

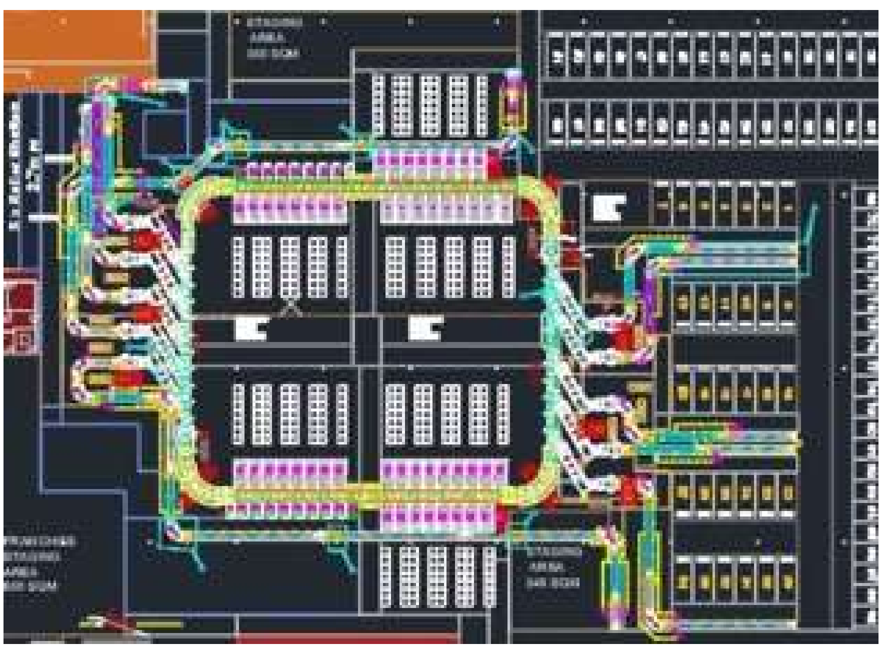
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

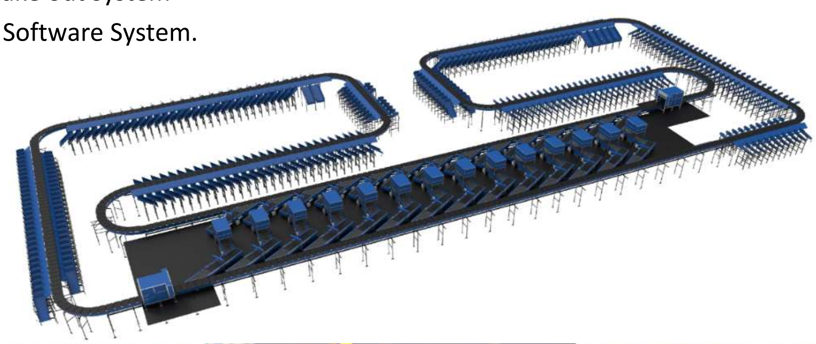
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

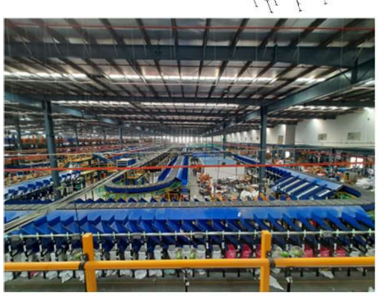
Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.





5.6 Project 6- (Ecommerce Client, India)

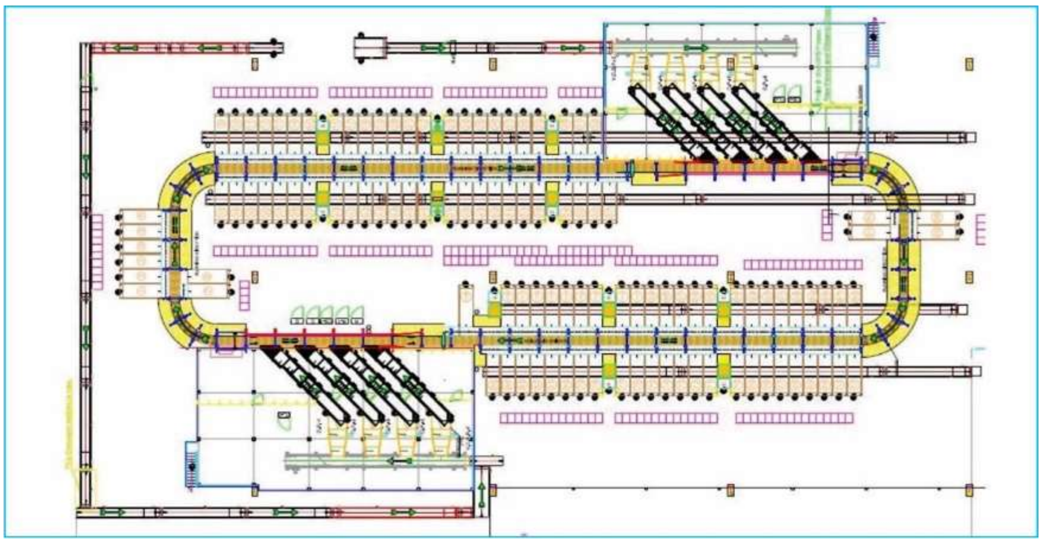
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.7 Project 7- (CEP Client, India)

India's largest Automated Parcel Sortation Centre is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

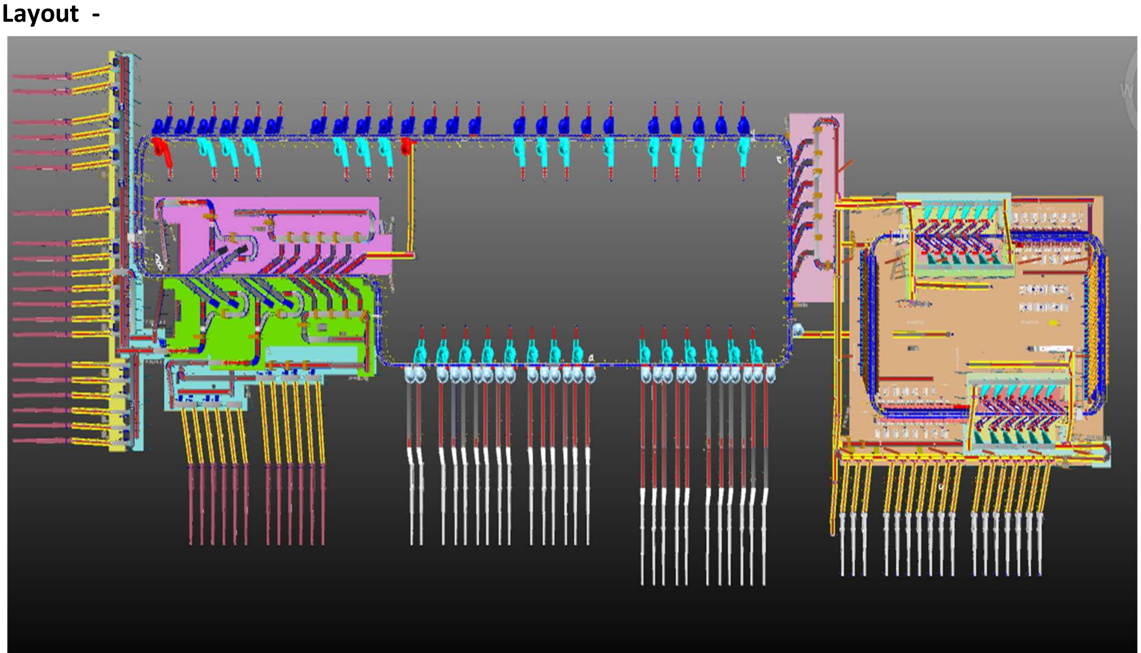
Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- 2 Sets of Double Decker CBS Sorters
- Mezzanine Structures (one large mezz on to install a double decker Sorter).
- Automated Singulators
- Fully Automatic Inductions
- Semi-Automatic Inductions
- Telescopic belt conveyors
- PVC Belt Conveyors
- Modular Belt Conveyors
- Spiral Chutes with Braking rollers
- 5-Sided Scanning Tunnels
- High speed weighing conveyors
- Direct Bagging Chutes
- Put to Light Chutes
- Volume Distribution systems
- High Availability Server Systems
- WCS

Layout -



6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detailed.

Falcon purposes to use its "Dual Belt Cross Belt Sorter" to provide the maximum benefits to Prime IQ in-terms of handling large shipment sizes and weight.

6.1 Shipment size loadable on the sorter

Falcon's Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	1000
Max Width	mm	600
Max Height	mm	500
Max Weight	Kg	10/Belt
Min length	mm	100
Min Width	mm	100
Min Height	mm	5
Min Weight	gm	100

Note- 10% of Semi-large shipment will be bifurcated visually & direct sent to ARM based outbound sorter

6.2 Shipments to be loaded on Sorter shall have the following characteristics:

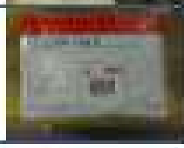
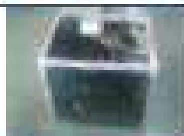
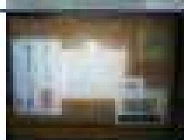
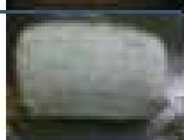
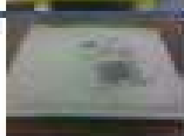
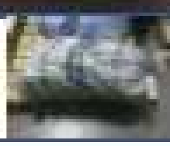
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Sortable Shipments

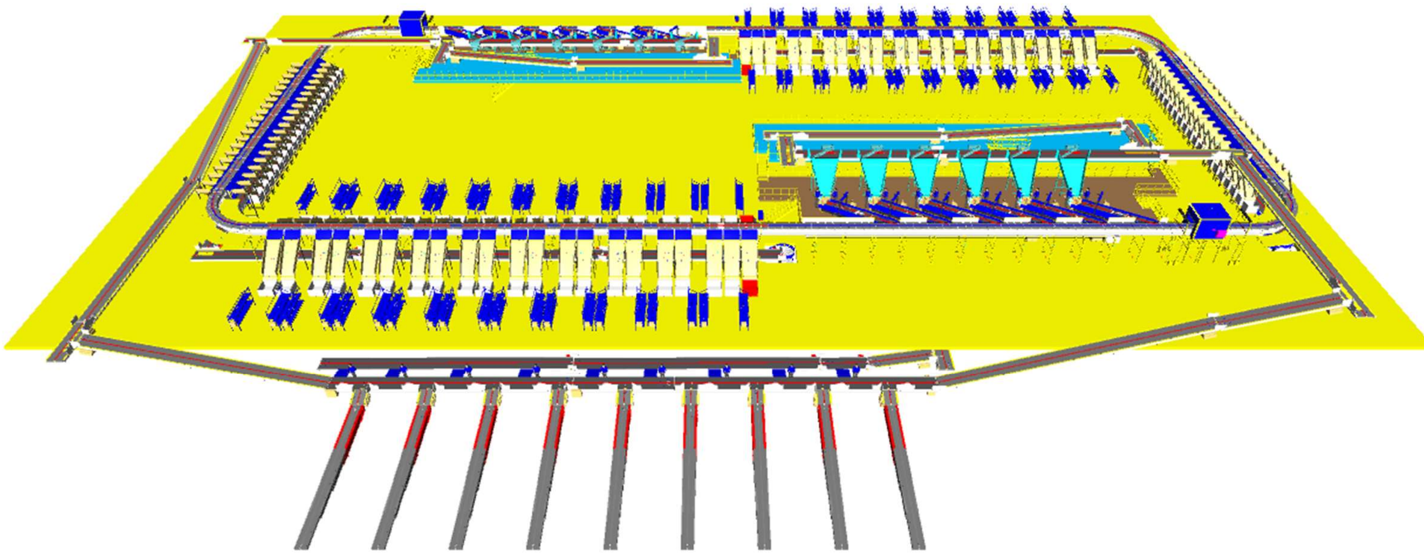
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting shipment as per Prime IQ's logic.

7.2 Summary of the System (G+1)



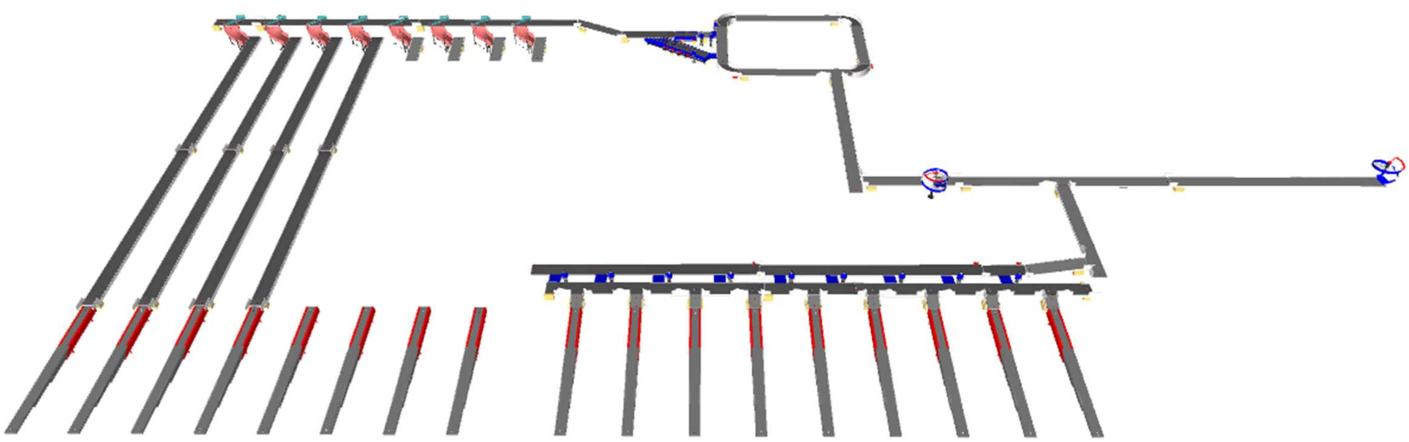
Process flow of the Loop CBS System-

1. **Infeed System: -** Shipments enclosed in bags are unsealed, and the shipments are dumped in bulk onto the telescopic belt conveyor, Semi large shipments are bifurcated visually & sent directly to arm sorter. CBS worthy shipments further transported to infeed lines of both VDS system. Subsequently, the bulk shipments ascend to a higher level and arrive at the VDS system. Once the shipments are within the VDS loop, they are evenly distributed among all inducts. In this solution one of the induct zone of each Loop is linked directly from below level via Infeed conveyors.
2. **Inducts: -** After the shipments are collected in the VDS chute, an operator picks and positions each shipment on the induct line, ensuring that the shipment is properly aligned and that its barcode is facing upwards. The feedlines then automatically induct the shipments onto the Cross-Belt Sorter Loop.
3. **Loop CBS:** - Once the shipments have entered the main loop, the Cross-Belt Sorter (CBS) efficiently sorts the shipments into their respective output chutes by utilizing the data provided by Prime IQ's sorting logic.
4. **Output Chutes: -** The shipments are discharged into two types of chutes.

a. **Secondary Chute (L-type) -** Within a zone of CBS system, there are a total of 40 collection-type chutes. Shipments collected within these secondary chutes undergo further sorting through the Put to Light Set up.
b. **Direct Bagging Chute (L-Type) –** Within a zone of loop CBS, there are total 40 direct bagging chute. Shipment collected within these direct bagging chutes are sealed with bag & transferred to bag take away conveyor.
c. **Rejection Chute-** Two rejection chute is present in a zone to handle rejected Shipments.

5. **Bag Takeaway Conveyor: -** Following the Direct bagging & secondary sorting process, the shipments are placed into bags and then manually loaded onto a bag takeaway conveyor located beneath the CBS loop. This conveyor transports the bags to a lower level below mezzanine.

7.3 Summary of the System (G Floor)



Process flow of the Outbound Bag Sorter System-

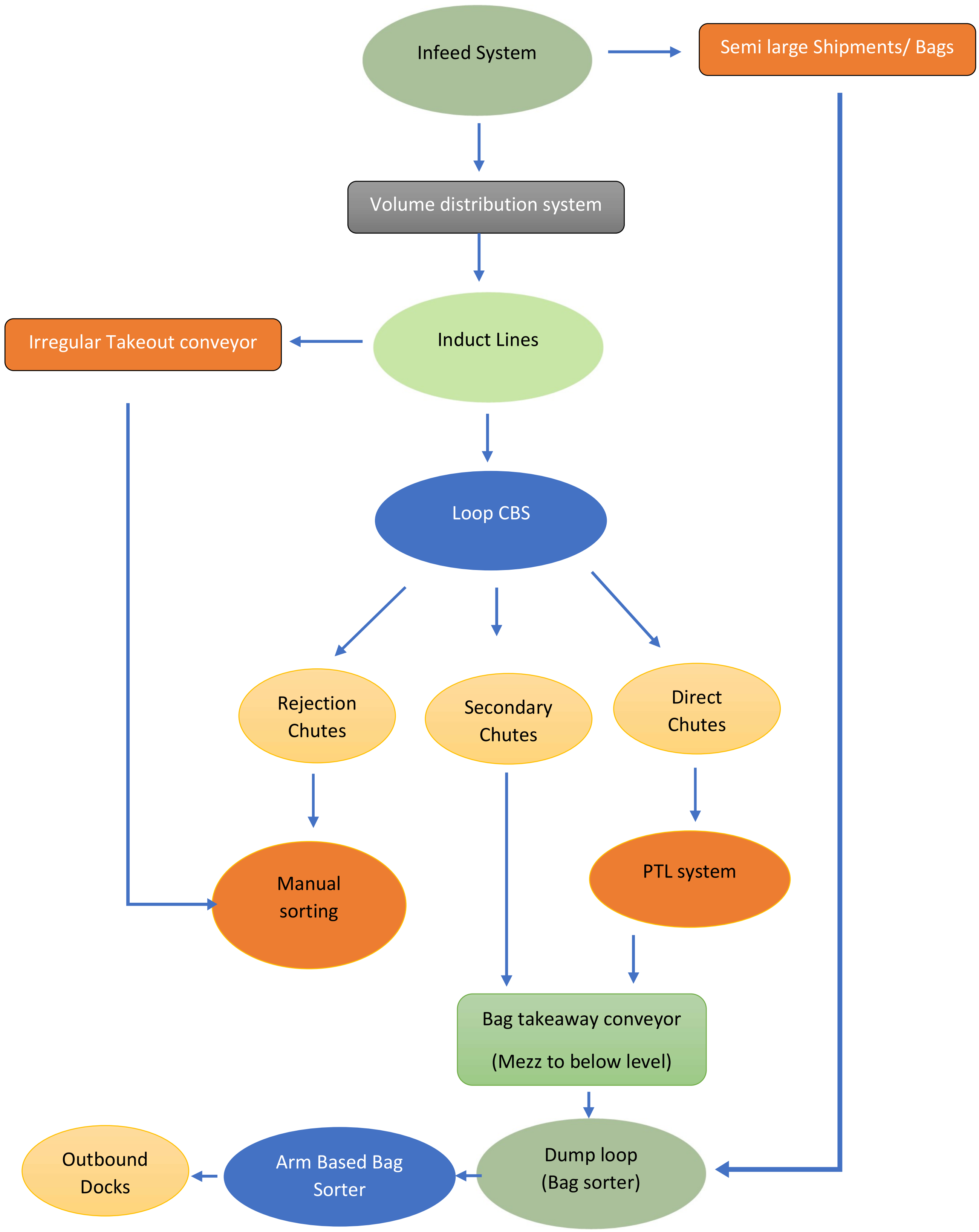
1. **Infeed System: -** Shipments enclosed in bags coming from above mezzanine level via two spiral chutes and the cross-dock bags coming from inbound telescopic belts are merged on a common highway conveyor that transport bags from all three points to VDS loop of Bag sorter.
2. **Inducts: -** After the bags reaches dump loop of bag sorter, operators manually pull the bags on two available inducts, scan the barcode via handheld scanner & then the bag proceed for final sortation.
3. **ARM Sorter:** - Once the bags have entered the arm sorter, It sorts the bags into designated outbound lane.
4. **Outbound Staging: -** Four lanes are not provided with PVC belts in order to maintain outbound staging area. Once the vehicle is ready for loading bags are manually placed on TBCs & hence loaded into particular vehicle.

7.4 Main benefits of the proposed solution

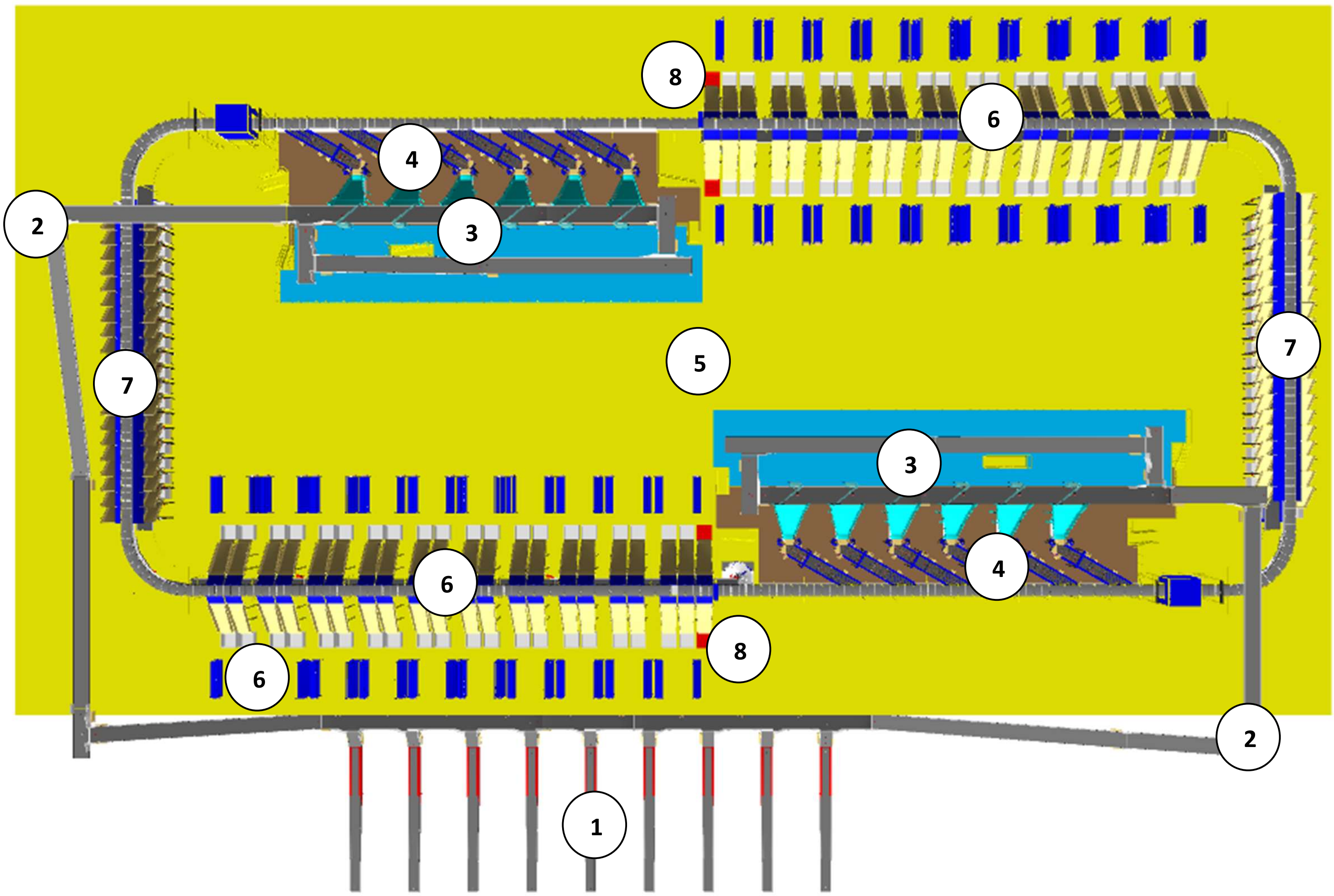
The proposed solution has the following advantages:

1. High operational throughput
2. Good Shipments conveying quality for the entire range of conveyable Shipments with a sorting conveyor speed of up to 2.1 m/s.
3. Low occupancy of floor space in the building.
4. Narrow discharge centers for the increased number of splits in limited space.
5. FALCON’s CBS can adapt to changing business requirements by adjusting its speed. to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Concept Description



9. Layout View (G+1)



Legend

1 – Telescopic Belt Conveyor

2 – Infeed lines

3 – VDS system

4 – Induct lines

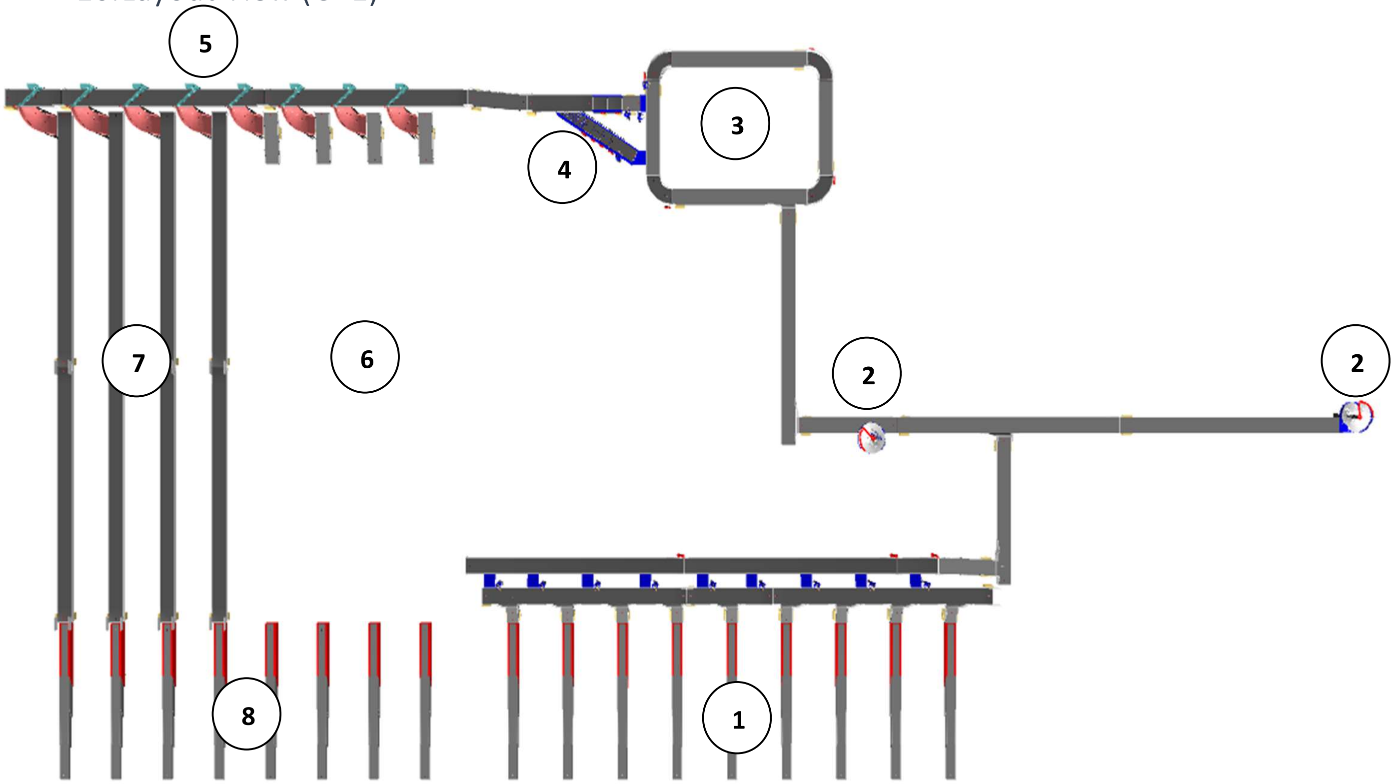
5 – Main CBS loop

6 – Secondary chutes (L-type)

7 – Direct bagging chute

8 – Rejection chutes (red colour)

10. Layout View (G+1)



Legend
1 – Inbound TBCs
2 – Spiral chutes (Bags)
3 – Dump loop VDS
4 – Induct lines
5 – Main Bag sorter
6 – Outbound staging Area
7 – Direct loading lanes
8 – Outbound TBCs

11. Proposed System Capacity Calculations

11.1 Sorter System Capacity

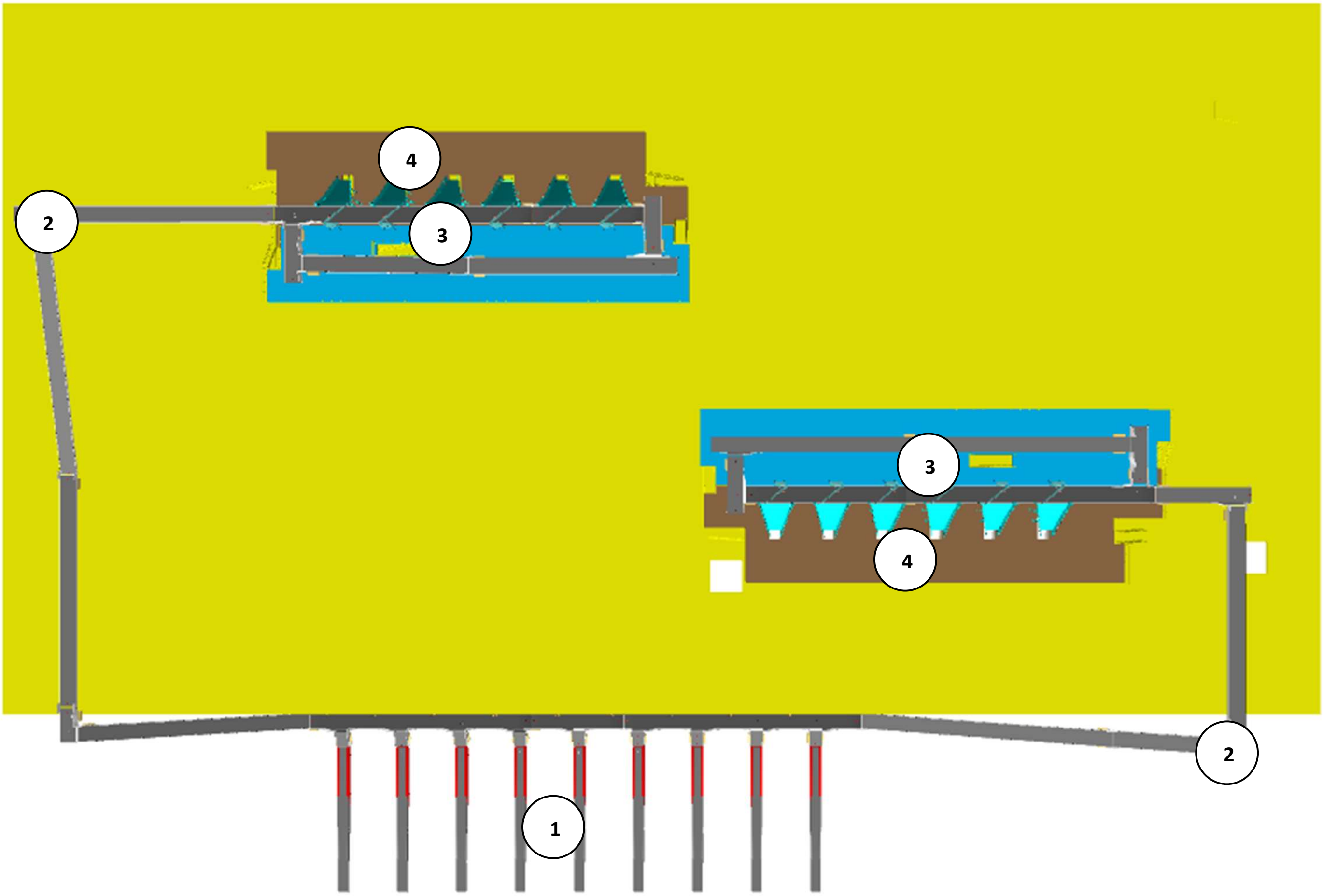
The following table shows the throughput calculation for the sortation system designed based on Prime IQ RFP requirements.

SPECIFICATION	VALUE
Sorter Type	**Loop CBS**
Speed	**2.1 m/s**
Pitch	**1,175 mm**
Carrier per hour	**6,434 CPH**
Belts per hour	**12,868 BPH**
Effective Designed Throughput of 1 loop CBS (A)	**25,372 Shipments per hour**
Per Feedline Capacity	**2,000 Shipments per hour**
No of Feedlines (Both Zones)	**12 Nos.**
Total Feedline designed capacity (B)	**24,000 Shipments per hour**
System designed throughput (Lower of A&B)	**24,000 Shipments per hour**
Designed throughput from one loops	**24,000 Shipments per hour**

Note- 10% of Semi-large Shipment above 600X600X600 mm will be separated & directly sent to ARM based sorter.

12. System description

12.1 Infeed System



Legend

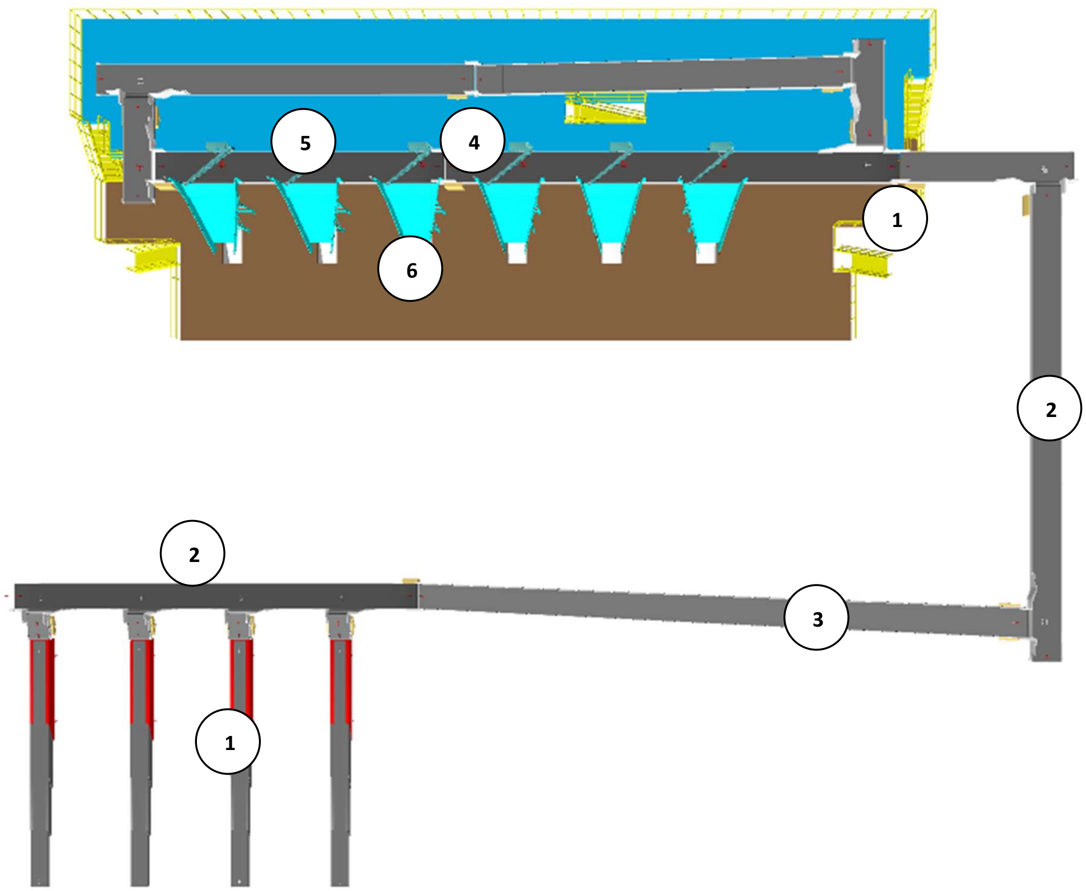
1- TBCs
2- VDS Infeed Conveyor
3- VDS Loop
4- VDS Chutes

CBS loop is equipped with set of PVC conveyors, facilitating transportation of bulk shipments from the Below mezzanine level into the VDS loop.

12.1.1 Infeed System

Infeed conveyor system consists of below components:

1. Telescopic belt conveyors
2. Straight PVC Conveyor
3. Inclined PVC Conveyor
4. Modular Conveyor
5. ARM Diverters
6. VDS Chute



12.1.1.1 Telescopic Belt Conveyor

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Base Length	Mm	6000
Extended Length (Total Length)	Mm	18000
Conveyor Belt Running Speed	m/min	18-30
Motor Rating	Type	0.75 Kw, 415V AC, 50 Hz
Load Capacity	Kg/m	50
Communication	Mm	2 x RS243, 1 x USB2.0, Ethernet

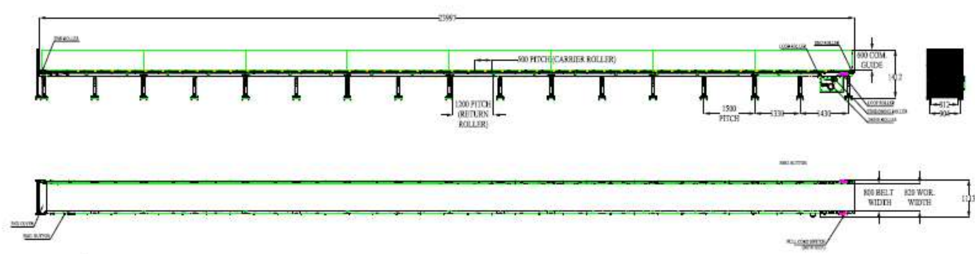
12.1.1.2 Straight and Inclined Powered Belt Conveyor

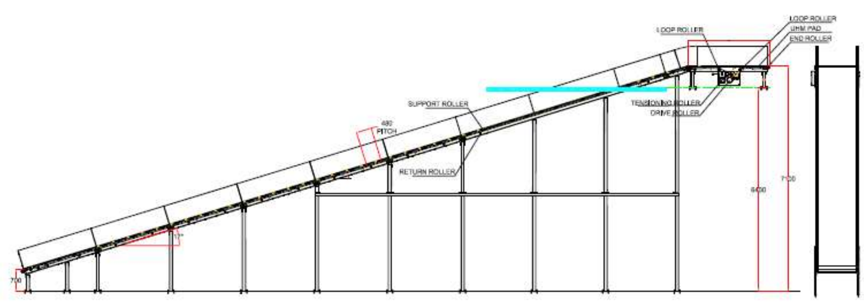
Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



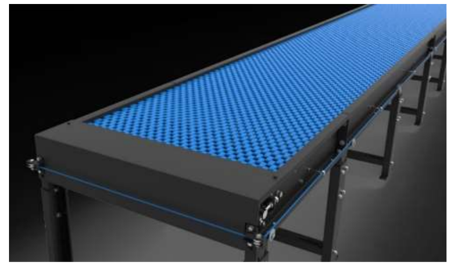


Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

12.1.1.3 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:

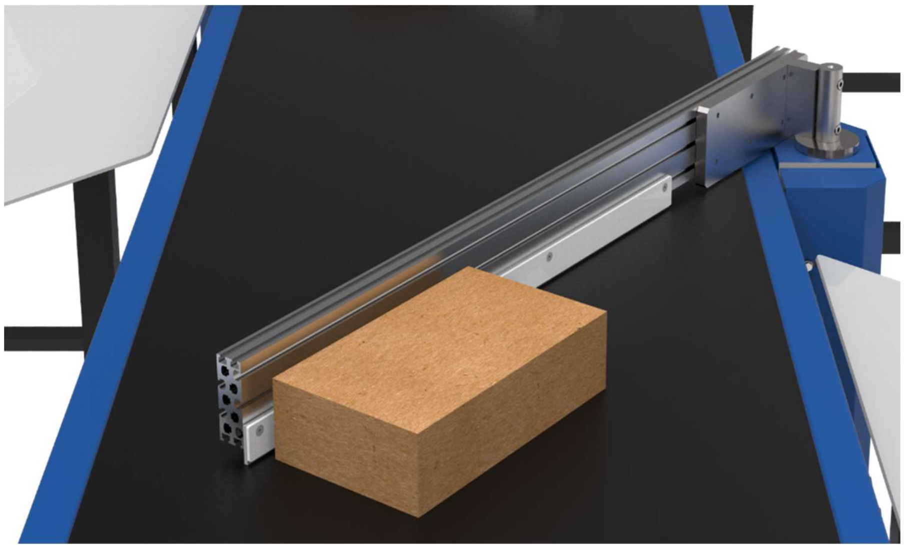


- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make	Make	Movex
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Wear Strip Material	MOC	UHMW PE1000
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

12.1.1.4 Diverter ARM

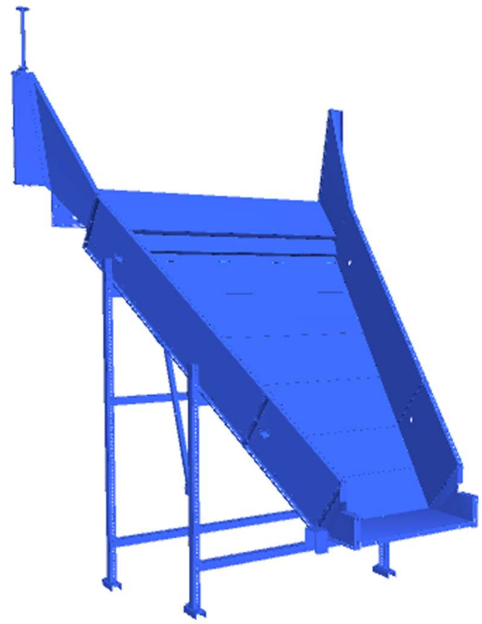
Linear Side Arm Diverter System of Falcon is an extremely cost-effective Sortation system designed for low to medium Volume Centres. Linear side arm diverter's simple and modular design gives it advantage of low cost, space saving, and modularity of increasing sorting points in future.



Specification	3.0 ARM XL
Property	Value
Actuation method	Electric induction motor
Belt type	Flat top modular
Belt width (mm)	1200
Arm length (mm)	2400
Max. product weight (kg)	40

12.1.1.5 VDS Chutes

A VDS chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level (at an operator level). It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area.



Calculation for Design: We have considered an average parcel size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average parcel size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type Chute	3.3	1.5	0.4	2.07

Assuming a 60% occupancy rate, about 45 **parcels** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Three Color Tower Light

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status

12.1.2 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	PVC Belt Conveyor	4810	4810	4.3	1210	1
2	PVC Belt Conveyor	4810	4810	4.3	1210	1
3	PVC Belt Conveyor	4810	4810	4.3	1210	1
4	PVC Belt Conveyor	NA	NA	4.3	1210	1
5	PVC Belt Conveyor	NA	NA	19.7	1210	1
6	PVC Belt Conveyor	NA	NA	3.8	1210	1
7	PVC Belt Conveyor	NA	NA	18.2	1210	1
8	PVC Belt Conveyor	NA	NA	7.2	1210	1
9	PVC Belt Conveyor	NA	NA	19.7	1210	1
10	PVC Belt Conveyor	NA	NA	4.3	1210	1
11	PVC Belt Conveyor	800	800	19.7	1010	1
12	PVC Belt Conveyor	800	800	5.3	1010	1
13	PVC Belt Conveyor	4560	4560	15.8	1210	1
14	PVC Belt Conveyor	5310	5310	4.3	1210	1
15	PVC Belt Conveyor	800	800	19.7	1010	1
16	PVC Belt Conveyor	800	800	5.3	1010	1
17	PVC Belt Conveyor	4560	4560	15.8	1210	1
18	PVC Belt Conveyor	4450	5604	12.3	1210	1
19	PVC Belt Conveyor	4450	5604	12.3	1210	1
20	PVC Belt Conveyor	NA	NA	19.7	1210	1
21	PVC Belt Conveyor	NA	NA	1.2	1010	1
22	PVC Belt Conveyor	NA	NA	1.2	1010	1
23	PVC Belt Conveyor	NA	NA	1.2	1010	1
24	PVC Belt Conveyor	NA	NA	1.2	1010	1
25	PVC Belt Conveyor	NA	NA	1.2	1010	1
26	PVC Belt Conveyor	NA	NA	19.4	1210	1
27	PVC Belt Conveyor	NA	NA	1.2	1010	1
28	PVC Belt Conveyor	NA	NA	17.2	1210	1
29	PVC Belt Conveyor	NA	NA	8.1	1210	1
30	PVC Belt Conveyor	NA	NA	1.2	1010	1
31	PVC Belt Conveyor	NA	NA	1.2	1010	1
32	PVC Belt Conveyor	NA	NA	1.2	1010	1
33	Modular Conveyor	8860	8,860	19.6	1238	1
34	Modular Conveyor	8860	8,860	8.3	1238	1
35	Modular Conveyor	8860	8,860	18.9	1238	1
36	Modular Conveyor	8860	8,860	12.1	1238	1
37	Telescopic Belt Conveyor					9

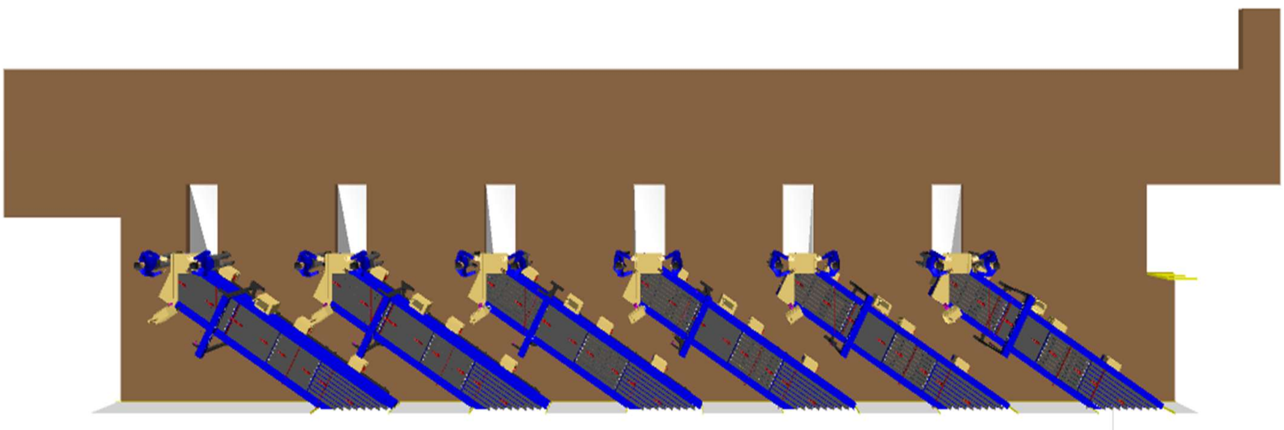
12.2 Induction to Sorter

The main purpose of the feedlines is to transport the shipments from the infeed points to the sorter by means of straight and curved belt conveyors, 30° merging conveyors, roller aligner conveyors and 30° induction lines. The flow of shipments on the belt conveyors is controlled by sensors in such a way that a gap is ensured between the shipments.

After the entry points to the unloading areas, all feed lines rise above the ground on mezzanine, leaving a working area underneath. The inclination of the belt conveyors towards the sorter will be maximized to obtain the best possible use of the space below the sorting loop. Shipment slippage will be prevented by using a maximum tilt angle of 15°.

Scales installed on the feedlines will acquire the weight of each shipment and will perform weight control to detect out-of-spectrum shipments.

In order to minimize the number of infeed lines while respecting the necessary flow rates for all unloading areas, the flows from the unloading stations will be merged, taking into account the maximum flow rate of each injector. Thus, the system's design ensures that the maximum capacity of each feed, considered as the sum of the flows from the connected inductions, is always less than the planned induction capacity of 2000 shipments/hour.



Induction Zone

12.2.1 Feedlines

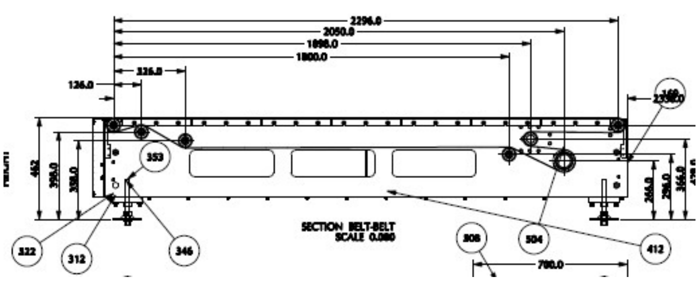
Each Loop CBS has a total 12 Automatic Feedlines with a Designed IPP of 2000 PPH.

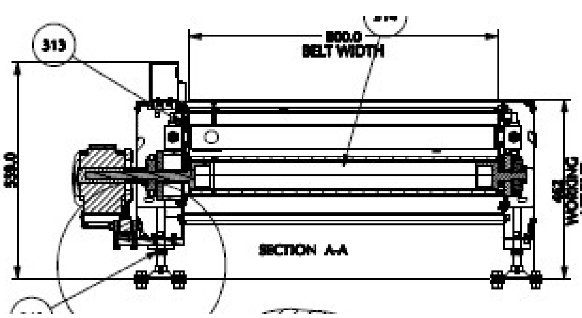
1. Each Feedline has Four Conveyor Modules-
   - Orientation Conveyor- 1 Nos.
   - Weighing Conveyor- 1 Nos.
   - Buffer conveyor- 1 Nos.
   - Intelligent Merge- 1 Nos.
2. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Shipment on the conveyor and prepare the Cross-Belt Cell for receiving the shipment with high precision.
3. Over the feedline, shipments are evenly spaced and smoothly transferred to the main Loop.

12.2.1.1 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.





12.2.1.2 Weighing Conveyor

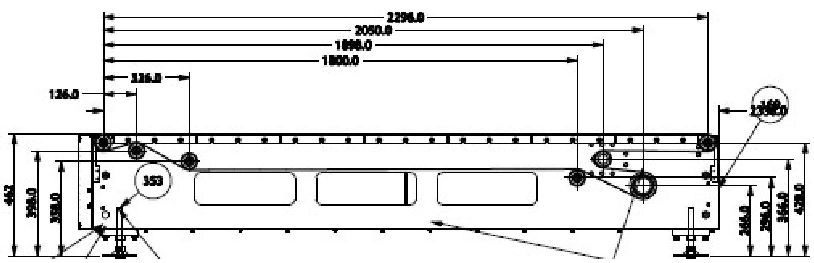
A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

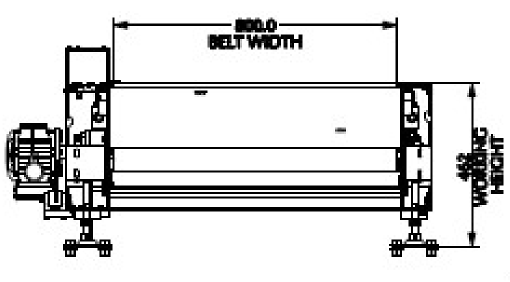
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Shipments.

Make- Bizerba/ Mettler Toledo

12.2.1.3 Orientation Conveyor

An Orientation conveyor is a type of conveyor system used to align & Release the products for induction onto CBS. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.

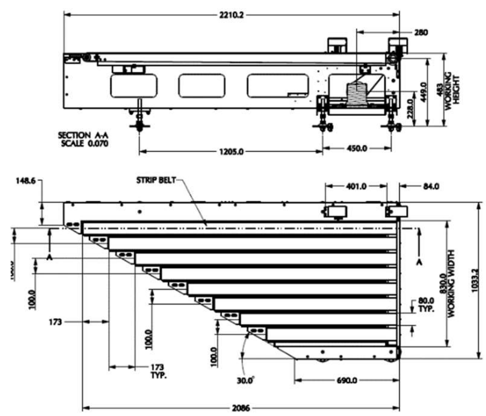




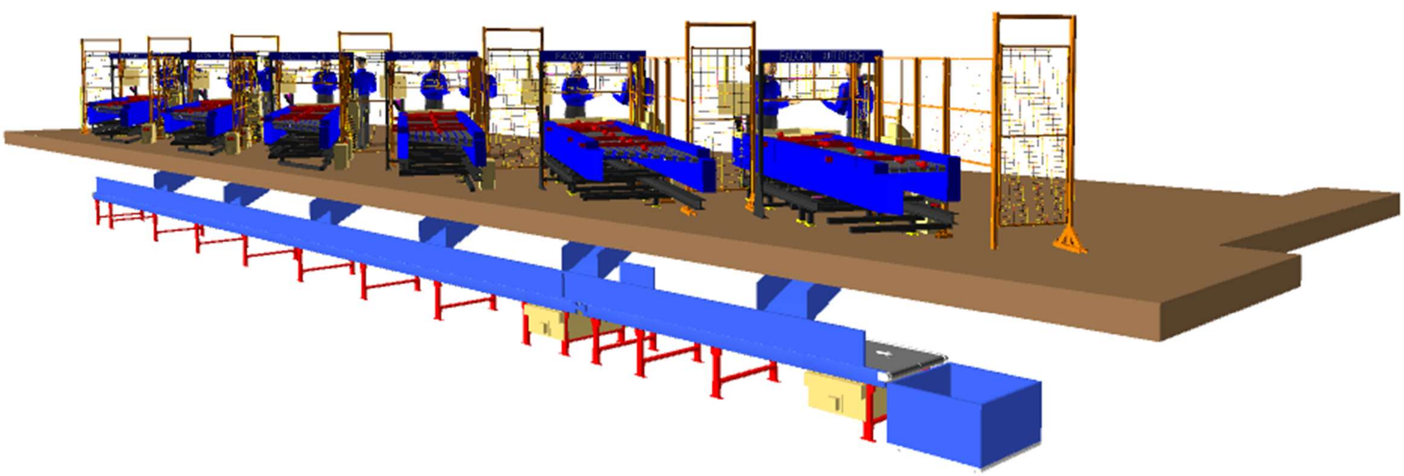
12.2.1.4 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

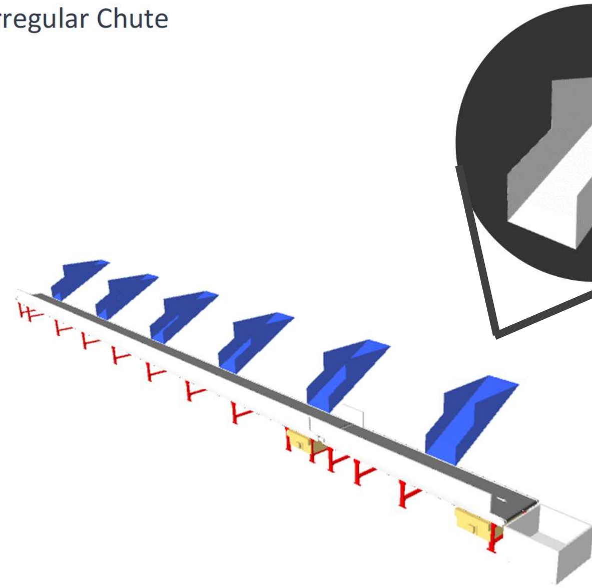
In the proposed solution this is 30° triangular Dual Belt high-speed conveyor used for inducting shipment/boxes directly on to the sorter. The Belts are Strip Belts for smooth shipment movement.



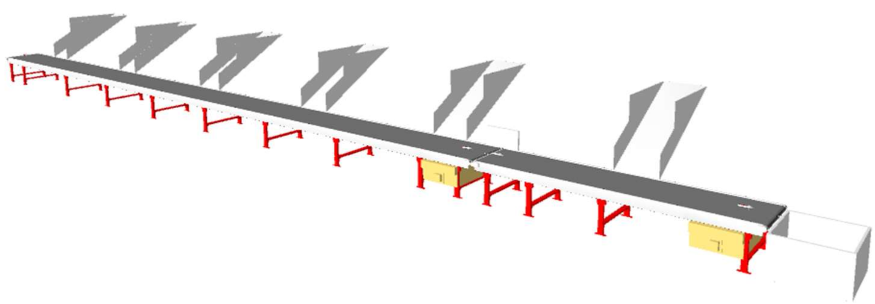
12.2.2 Irregular Takeout Conveyor



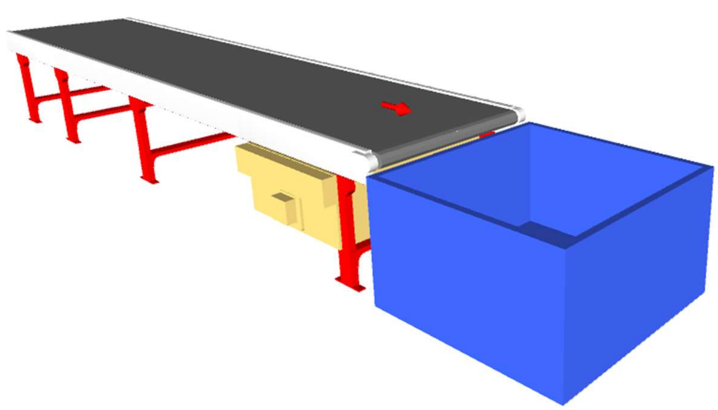
12.2.2.1 Irregular Chute



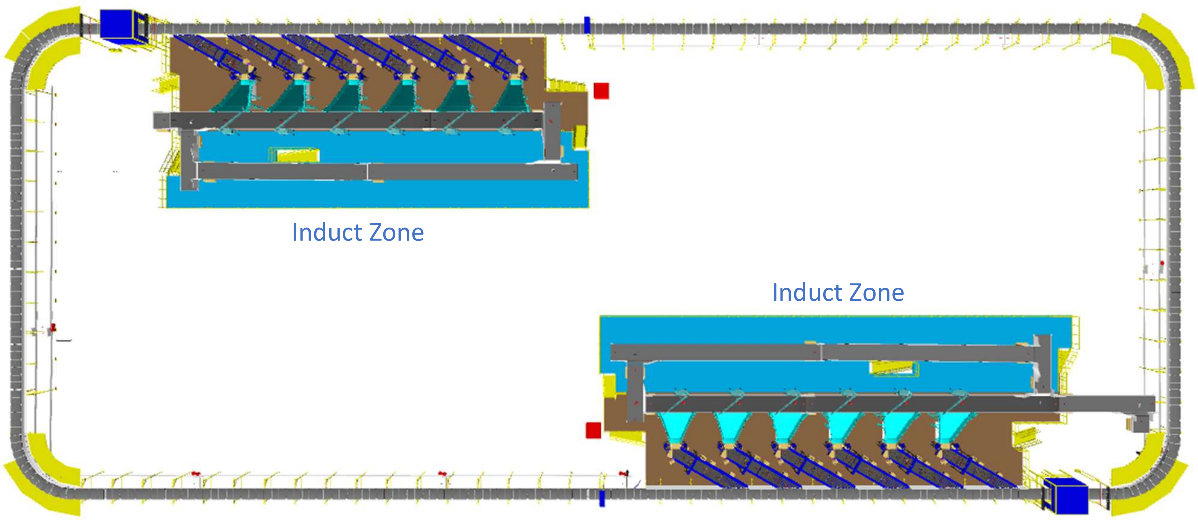
12.2.2.2 Irregular Takeaway Conveyor



12.2.2.3 Irregular Collection Bin (In Prime IQ's Scope)



12.2.3 Main Loop



1. The proposed Single Deck CBS Sorter will be installed on ground level. The Sorter runs at 3300 mm from ground level.

2. We have provided loop with dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.

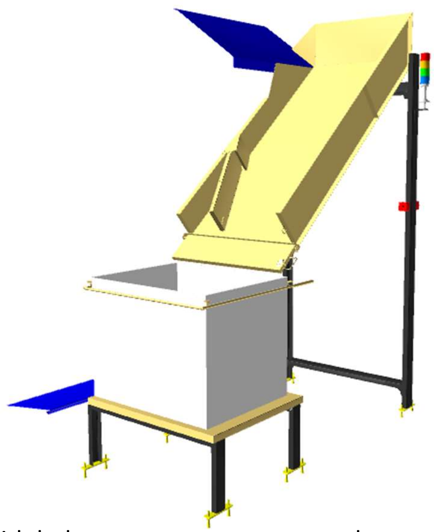
3. One loop CBS has a total length of 245 m & 209 number of carriers.

4. As Shipments pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches the respective chute, the carrier carrying the shipment for the chute will actuate and drop the shipment in the assigned chute.

12.2.4 Sorting Output

12.2.4.1 Direct Bagging Chutes

A direct bagging chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments into bags placed beneath the chute.



Each Chute is equipped with below components to ease the operations:

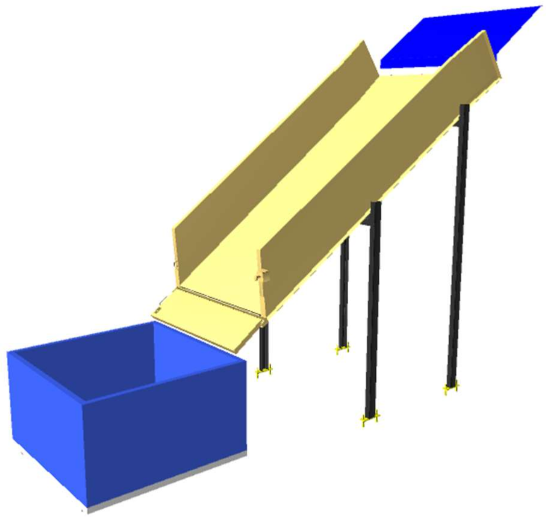
- Chute Full Sensor
- Three Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.2.4.2 Secondary chute

A collection chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & collect the shipments within its area once the end gate is closed.

Each collection type chute has capacity to accumulate 58 shipments, The sorted shipments will travel along these collection chutes and get accumulated at the bottom of the chute.



Calculation for Design: We have considered an average shipment size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type Chute	3.7	1.2	0.6	2.664

Assuming a 60% occupancy rate, about **58 shipments** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

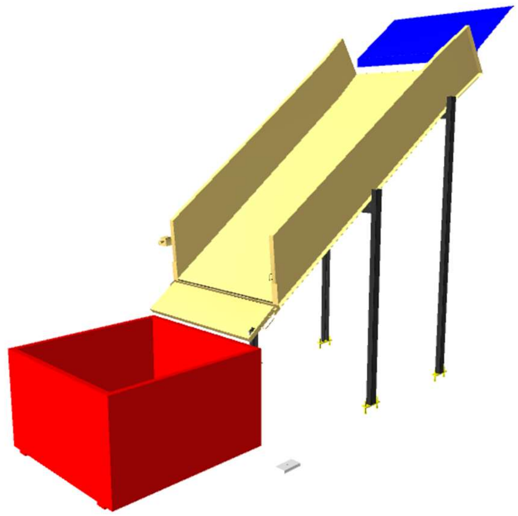
- Chute Full Sensor
- Three Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.2.4.3 Rejection Chute

Rejection Chute is a type of chute used for the smooth descent of materials or objects from an elevated position to a lower level. It utilizes base metal to guide and support the flow of items along the chute & end gate can be closed to collect the shipments within its area.

Rejection chute has capacity to accumulate 16 shipments, The sorted shipments will travel along these collection chutes and get accumulated at the bottom of the chute if the end gate is closed.



Calculation for Design: We have considered an average shipment size of 300 X 300 X 300 mm while designing the capacity of chutes.

Average shipment size considered for chute volume calculation	L(m)	B(m)	H(m)	Volume (m3)
	0.3	0.3	0.3	0.027
Collection type PTL Chute	2.0	0.9	0.4	0.72

Assuming a 60% occupancy rate, about **16 shipments** will readily fit into these chutes.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Three Color Tower Light
- Push Button

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status
Push Button	To Start/Stop Sorting Operations

12.2.4.4 Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keep on altering the PTL configuration to increase the system productivity and give you the extra competitive edge.

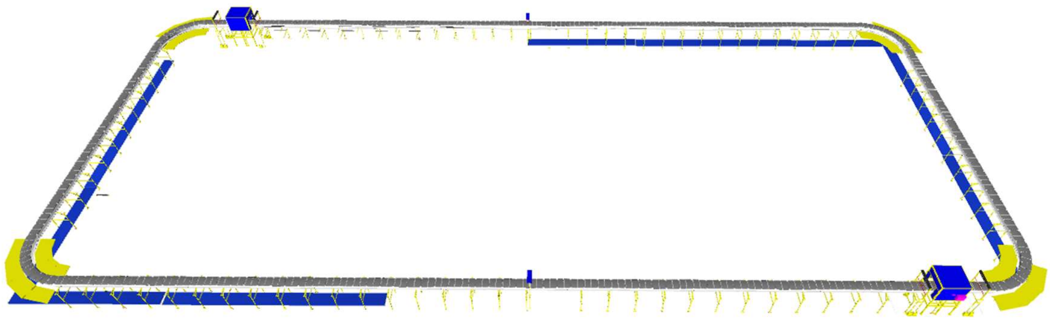
The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology provide high throughput with single operator with high accuracy

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

12.2.5 Bag Take away Conveyor

In each Loop CBS, there is a series of Bag Takeaway Conveyors, spanning a combined length of 223 meters. Following the secondary sorting process, bags are securely fastened and manually placed onto these conveyors, which are positioned below the Loop CBS to cover all chute locations. These conveyors then transport the collected bags back to the lower level.



12.2.5.1 Conveyor BOQ

S No.	Name	EL_1	EL_2	Length (m)	width (mm)	Set
1	Bagging Conveyor	550	650	14.7	1210	1
2	Bagging Conveyor	3800	3900	19.5	1210	1
3	Bagging Conveyor	3800	3900	19.9	1210	1
4	Bagging Conveyor	550	650	19.9	1210	1
5	Bagging Conveyor	550	650	19.9	1210	1
6	Bagging Conveyor	3800	3900	19.9	1210	1
7	Bagging Conveyor	3800	3900	13.7	1210	1
8	Bagging Conveyor	3800	4032	5.1	1210	1
9	Bagging Conveyor	3800	3900	19.9	1210	1
10	Bagging Conveyor	3800	3900	11.3	1210	1
11	Bagging Conveyor	3800	3900	19.9	1210	1
12	Bagging Conveyor	3800	3900	19.9	1210	1
13	Bagging Conveyor	3800	3900	19.9	1210	1

13. Description of Components of Equipment

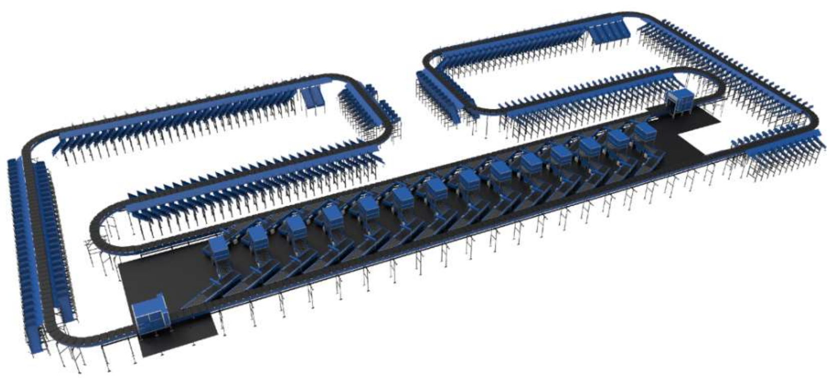
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

13.1 Cross Belt Sorter -

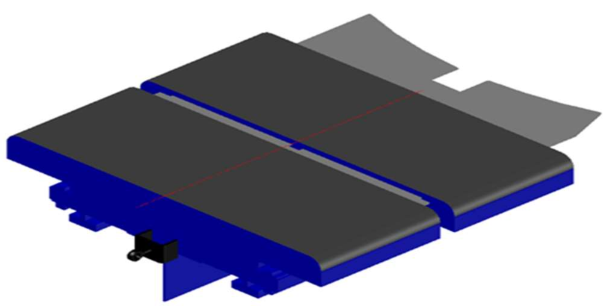
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



13.1.1 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable shipment recirculation.



13.1.2 Servo Roller

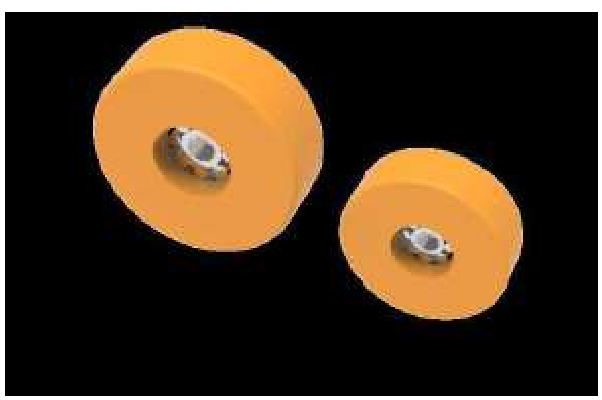
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

13.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

13.1.4 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle.



13.1.5 Non-contact based linear motor drive

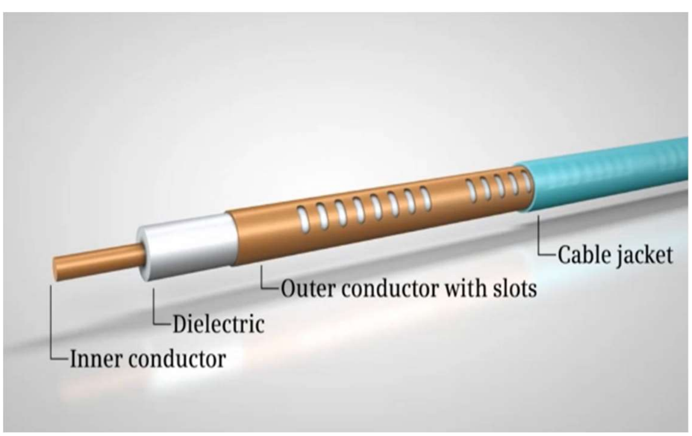
Extremely High Energy Efficient Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.

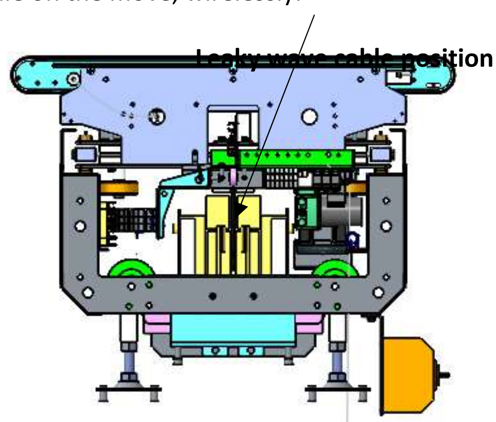
13.1.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

13.1.7 Data Transmission

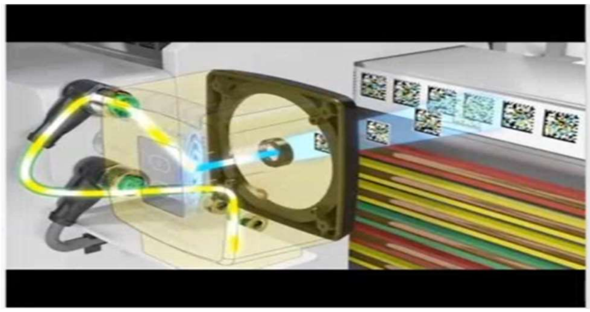
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.





13.1.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



13.1.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length	~245 m
Sorter Actuation Technology	Electric
Chutes	
Direct Chute	80
Secondary Chute	80
Rejection Chute	4
VDS Chute	12
Irregular Chute	12
Height of Sorter	~3300 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	209 Nos.
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Feedlines (Semi-Automatic)	12 Nos
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

13.2 Scanning & Sensing on Main Sorter Loop

13.2.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan shipment 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time. As per PRIME IQ requirements, Scanning is in PRIME IQ scope.

Falcon needs to supply Volume Measurement System. VMS 4200 quoted as option for Dimensioning of Shipments.

13.2.2 Shipment Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple shipment positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Shipment on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).

13.2.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the shipments on the empty carriers. (Camera based)

14. Proposed System Technical Details

14.1Mechanical equipment (CBS & Bag Sorter)

Pos.	Qty.	Description	Value
1	2	**Infeed System** • Powered Belt Conveyors • Modular Belt Conveyor • Telescopic belt Conveyor • Bag Takeaway Conveyor • XL ARM Diverters • VDS Chutes	 ~441 metres (48 Modules) ~59 metres (4 Modules) 09 Nos ~223 metres (13 Nos) 12 Nos 12 Nos
2	4	**Feedlines** Consists of • Receiving Conveyor • Weighing Conveyor • Buffer Conveyor • Intelligent merge	 1 Set 1 Set 1 Set 1 Set
3	2	**CBS Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch **Including:** • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Dimension Scanning System • E-Stops • Hooters • Fencing	 3200mm (from mezz level) ~245 m ~2.1 m/s Linear Motor 1175 mm
4	1	**Bag Sorter** **1 Arm based bag Sorter.** • XL ARM Diverters • Angle merge • Receiving Conveyor • Modular Conveyor • Buffer Conveyor • Modular turns 90 degree	 8 Nos 1 Nos 1 Nos ~94 meters (7 Modules) 5 Nos 4 Nos
4	2	Sorter Outputs • Direct bagging Chute • Secondary Chute • Rejection Chute • Irregular Takeout Chute	 80 Nos 80 04 Nos 12 Nos

5	**2**	**PTL System** • PTL Racks • PTL Modules • PTL Control Box • Handheld scanner	 800 Nos 800 Nos 80 Nos 80 Nos

14.2Electrical Equipment

Electricals			
1	**1**	**Consists of** Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

14.3Control System

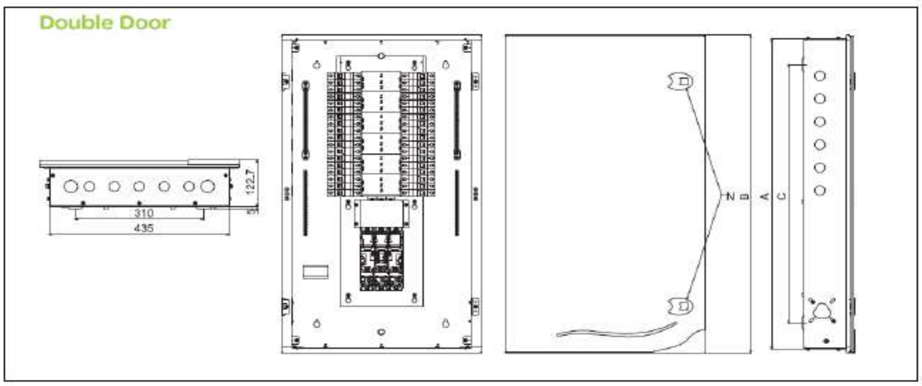
Components			
1	**1**	**Consists of** Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

15.Electrical System

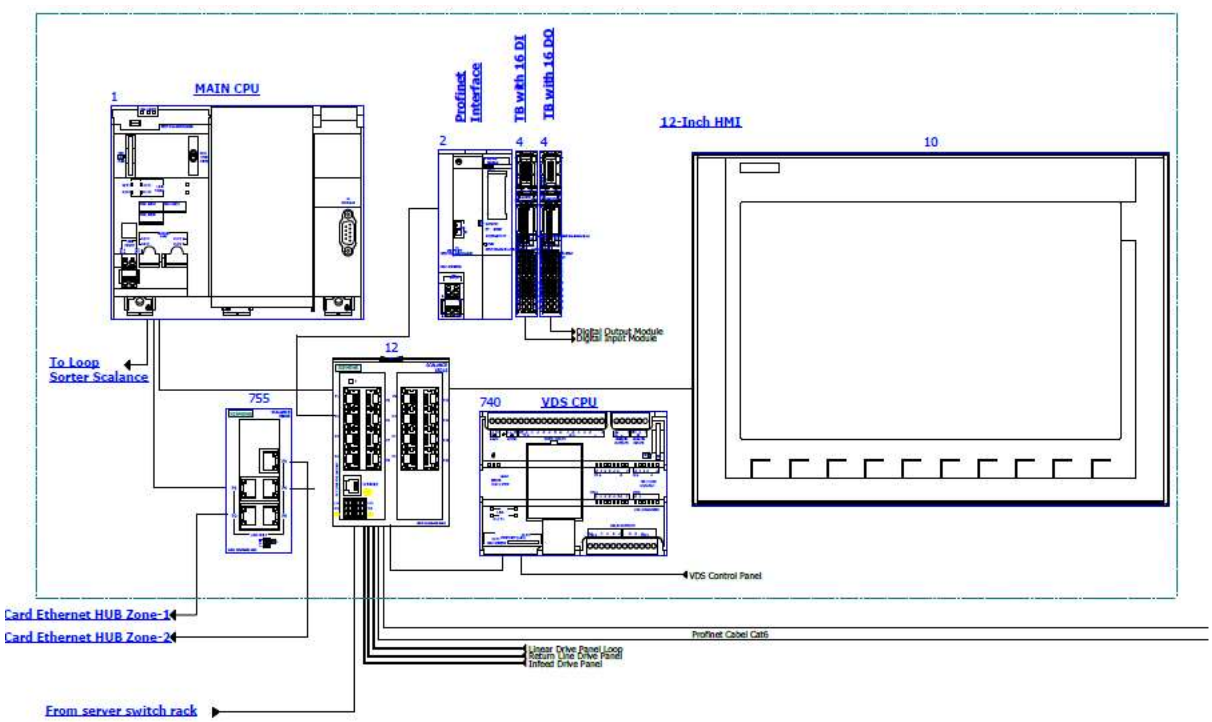
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

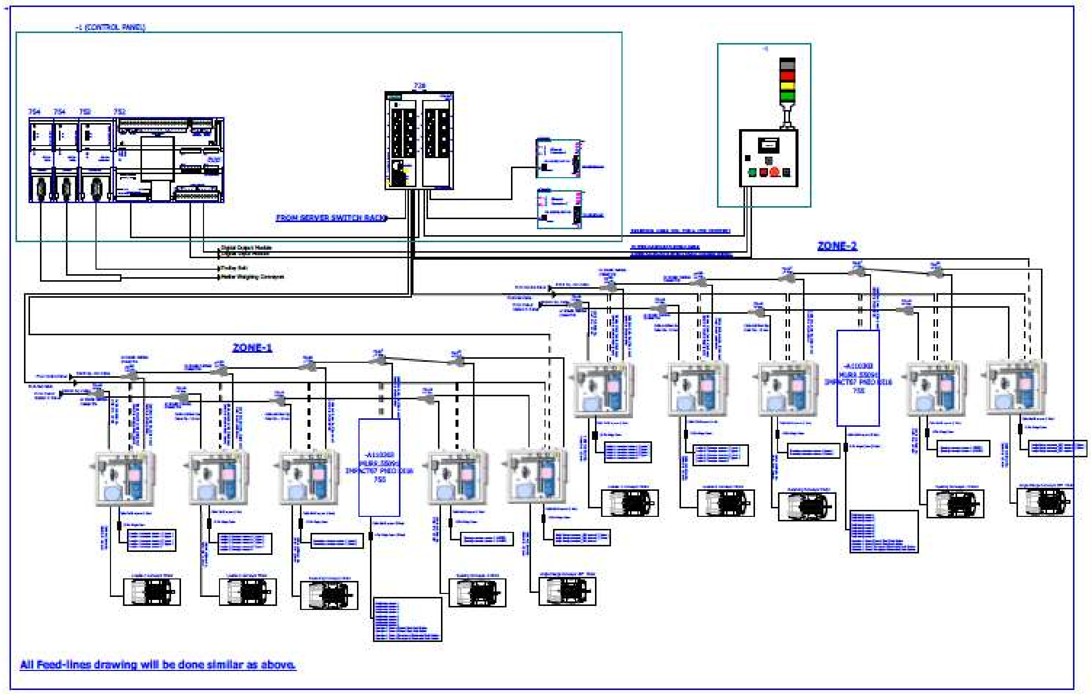
15.1 Reference Picture of Power Distribution Panel



15.2Main Control Panel (Reference)



15.3Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

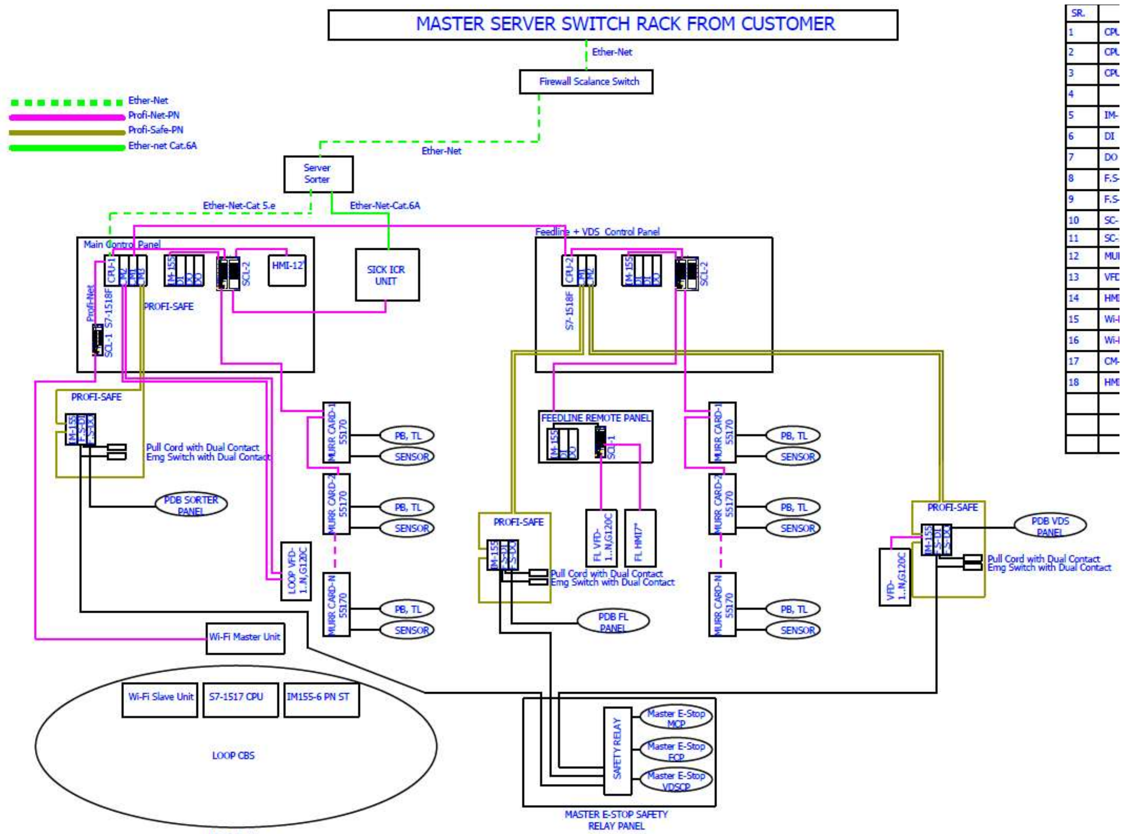
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

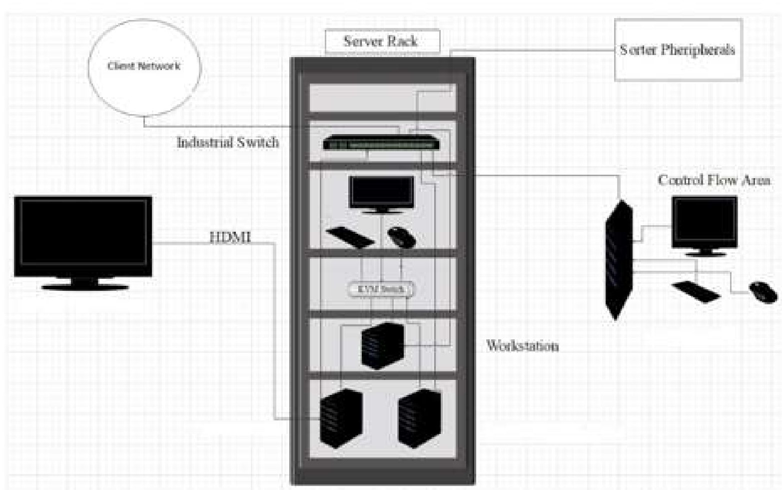
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

16. Controls Architecture (Ref.)



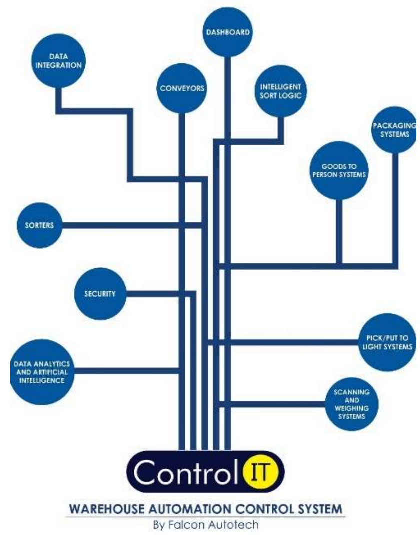
17. IT Architecture (Ref.)



17.1HAA Server (In Prime IQ Scope)

S.no	Description	Qty
	20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40	
1	**Synology_storage_DS723+** 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports, 2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays | 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	2
2	**Dell Tower Model T440** -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	2
3	**Dell PowerEdge T40** Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3 Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	1
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

18.Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

19.Falcon's Visual Inspection System (SCADA)

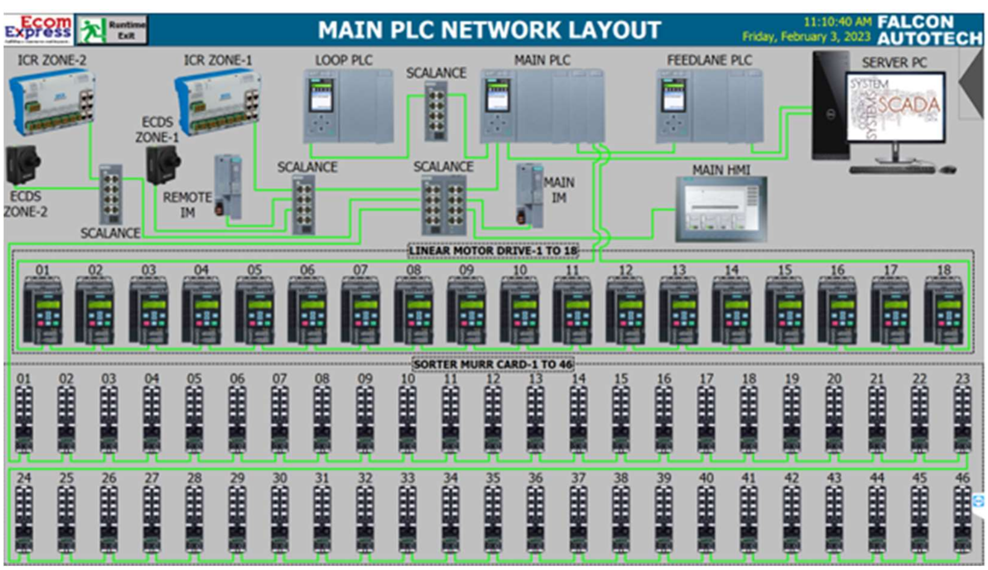
SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities.

The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for the Prime IQ Riyadh Hub. This SCADA system receives from each monitored sub-system all information on their operating status in real time.
At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

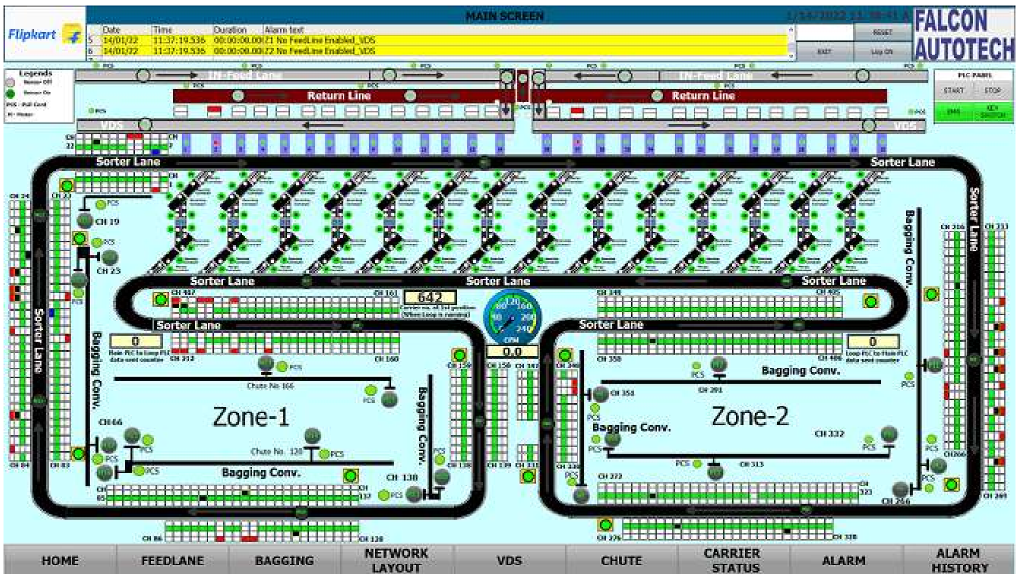
19.1FIELD DATA ACQUISITION

The field data acquisition function is performed by the SCADA system connected to the sorters' PLCs.
The communication with the PLCs is done using equipped CPU cards that are able to manage the communication with the PLC on the Industrial Ethernet network, without overloading the server.
The acquisition of signals from external systems is carried out via the PLC, using dry contacts.

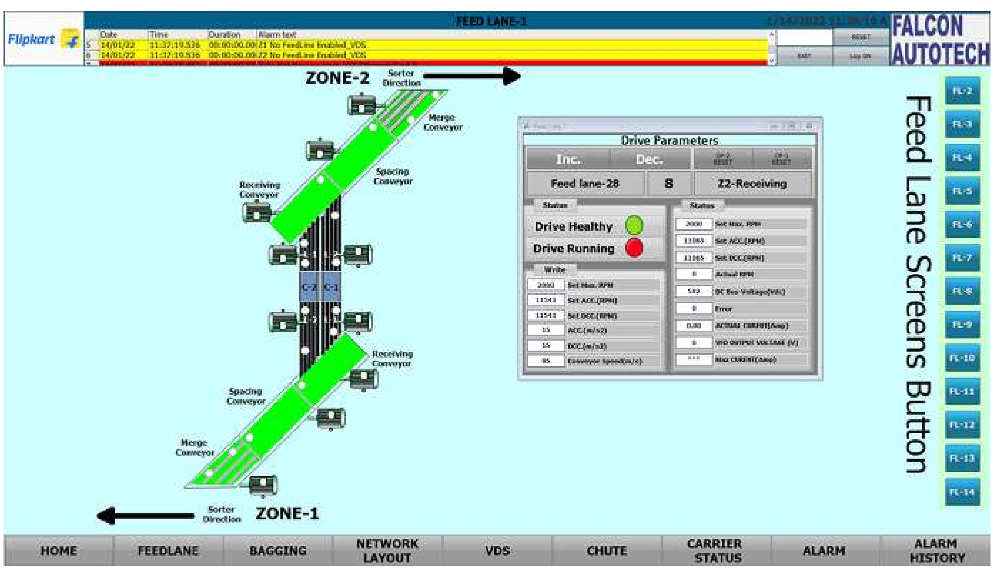


19.2ANIMATED SYSTEM VISUALIZATION

The animated view represents the dynamic graphical user interface that allows real-time monitoring of the controlled systems and the execution of their control procedures.
The various states of the digital signals are displayed on the screen by graphic symbols. All views are web based with responsive capabilities when required so that various devices can be used to display status and information on PC. The types of information that can be displayed on each type of device will be discussed during the design phase of the project.



Example of a synoptic view



19.3ALARM MANAGEMENT

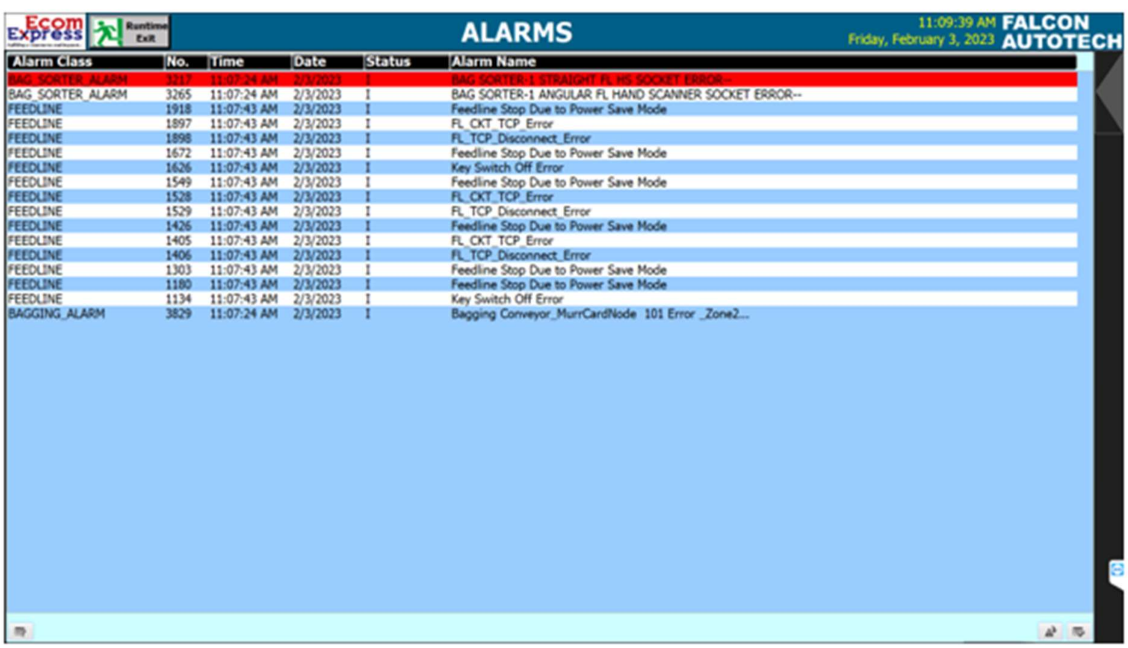
The alarm pages display a series of information to identify the nature of the alarm or event, the elements involved and the time.

The alarm management includes:

- The Alarm name.
- The date on which the above-mentioned alarm was activated.
- The moment the alarm goes off.
- The date on which the alarm is activated again.
- The moment the alarm is activated again.
- The sub-system that activated the alarm.
- The area where the alarm was triggered.
- The name of the beacon that activated the alarm.
- The alarm state.
- The alarm value.
- A complete alarm description.

The historic data of each alarm is stored in the SCADA database. This type of fault detected includes:

- ✓ Sensors.
- ✓ Drive Fault.
- ✓ Lack of monitored equipment.
- ✓ Etc.



Sample Alarm History Screen

20. Key Components Make

Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners (VMS 4200)	SICK
Weighing Scales	Bizerba/ Mettler Toledo
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron
Cables	LAPP/ Equivalent
Switch Gear	Schneider/Equivalent
Bearings	NTN/SKF/Equivalent
Power Transmission Systems	Vahle
HMIs	Siemens

21.Principle of Safety

21.1E-Stops

As per Falcon standard.

21.2Pull Cords Switch

As per Falcon standard.

22. Program Organisation

22.1Program/ Project Schedule

S.No.	Activity	Timeline
1.	Receipt of PO	Week 0
2.	Project Kick off	Week 1
3.	DAP Submission	Week 6
4.	DAP Approval	Week 8
5.	Detailed Designing	Week 13
6.	Manufacturing & Procurement	Week 20
7.	Inhouse Assembly	Week 25
8.	Dispatch	Week 27

**Above plan is tentative, detailed plan will be shared post order confirmation.*

22.2Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Prime IQ
3. Weekly/Fortnightly Meeting Prime IQ to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

23. Prime IQ Responsibility

23.1Prime IQ Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- Prime IQ to provide HPT/ Forklift for material movement for complete duration.
- The possibility of authorizing access to the site and the execution of the installation

work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.

- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.
- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

23.2Responsibilities of Prime IQ During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

23.3Prime IQ Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of PRIME IQ.

24.Commercial Terms

24.1Price Sheet

S. No	Component	Set	Price (USD)
1	Loop CBS System	1	$ 1,552,903
2	ARM sorter	1	$ 91,692
3	Conveyor Automation	1	$ 1,469,625
4	Output Chutes	1	$ 307,445
5	PTL System	1	$ 191,262
6	Bought-out items	1	$ 21,256
7	Software Packages	1	$ 16,923
8	Toolkit	1	$ 1,228
9	Installation & Commissioning	1	$ 655,566
10	Project Management	1	$ 92,728
Grand Total (USD)			**$ 4,400,626**

The Total Price is to be understood.

- **Packaging & Forwarding - complementary.**
- Freight - Excluded
- Taxes as applicable
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area is in customer scope

24.2Payment Terms

50% Advance Along with PO

50% Against material readiness (Before Dispatch)

**All invoices (if due as per payment terms) should be cleared within 7 days from the date of receipt of undisputed invoice by the buyer*

25. Warranty Period

Falcon's offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

26. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Server System**
- **Construction Power**
- **Steel Works (Base Mezzanine & Operator Platform)**
- **Collection Trolleys below chutes**
- **PC for SCADA.**
- **Fans at chutes & inducts.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.

